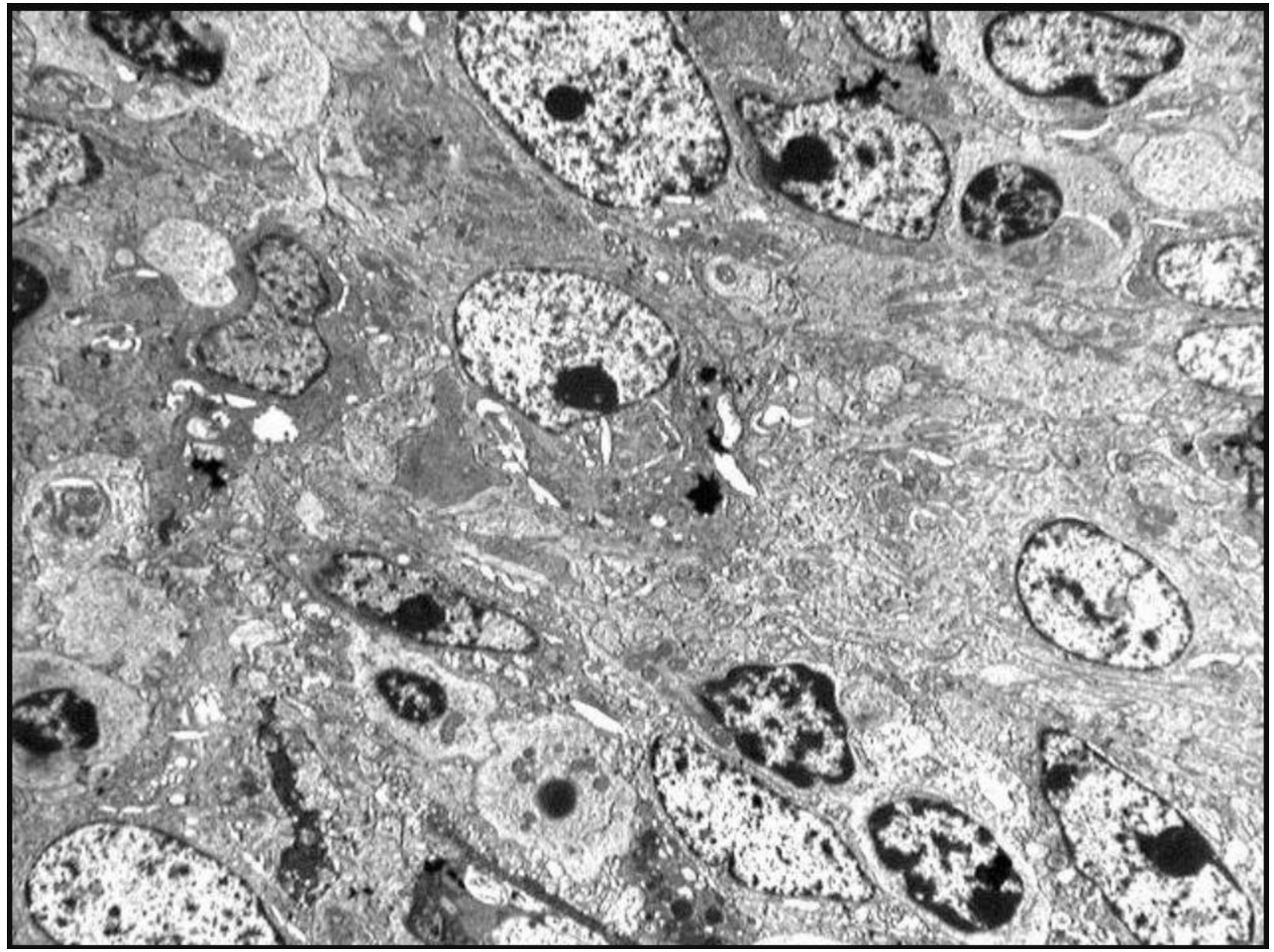
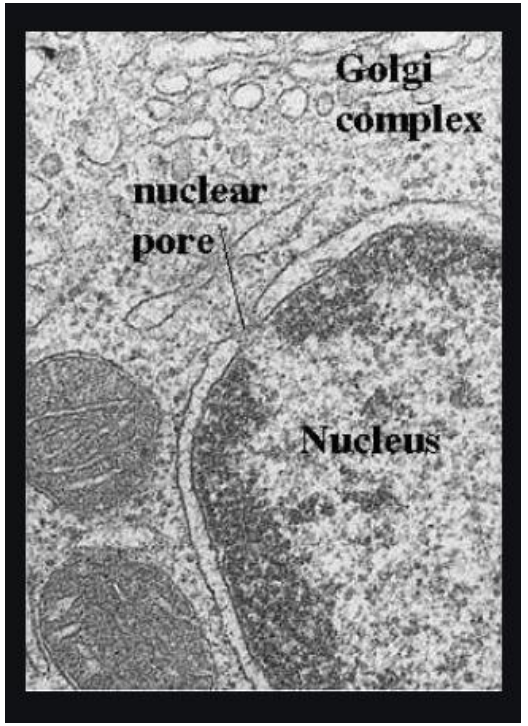
Guía análisis imágenes microscopía electrónica

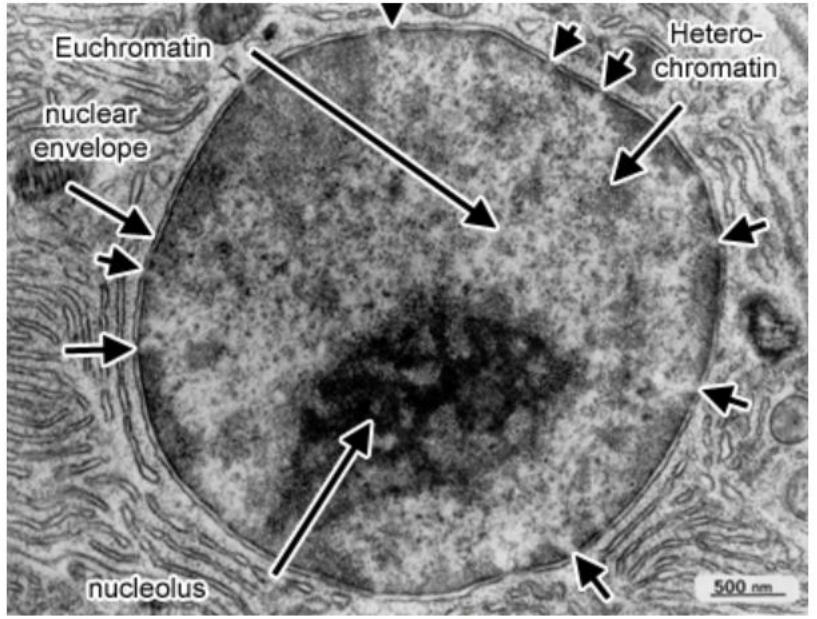
(material de referencia para las prácticas 8-9)

Vanja Tepavcevic Mandic/Isabel Martinez Garay

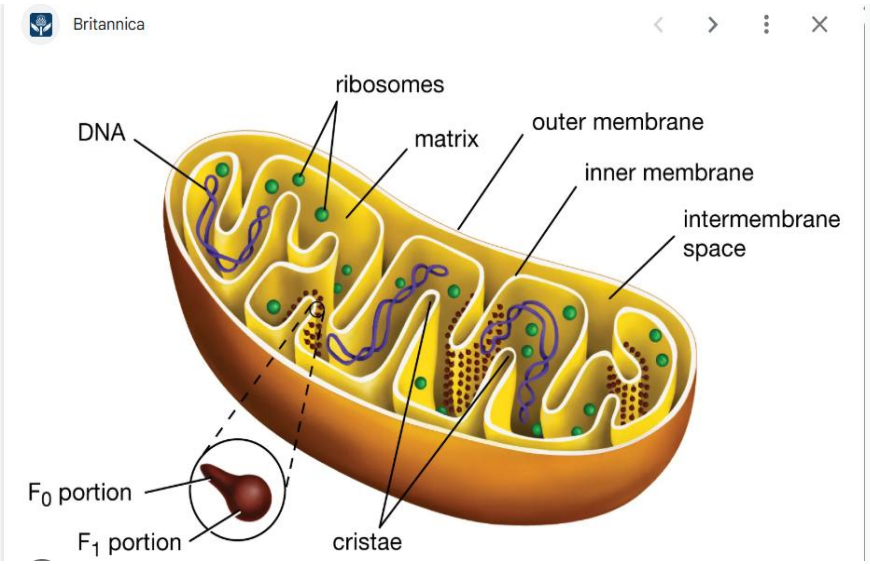
Núcleo, nucleolo, cromatina, poro nuclear

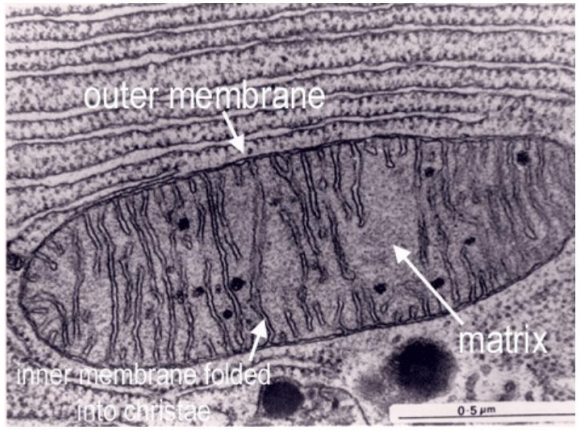


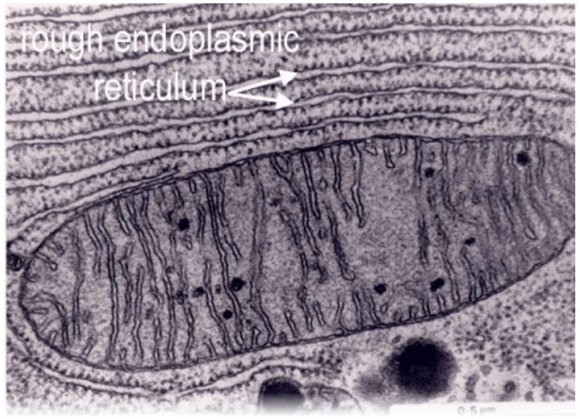


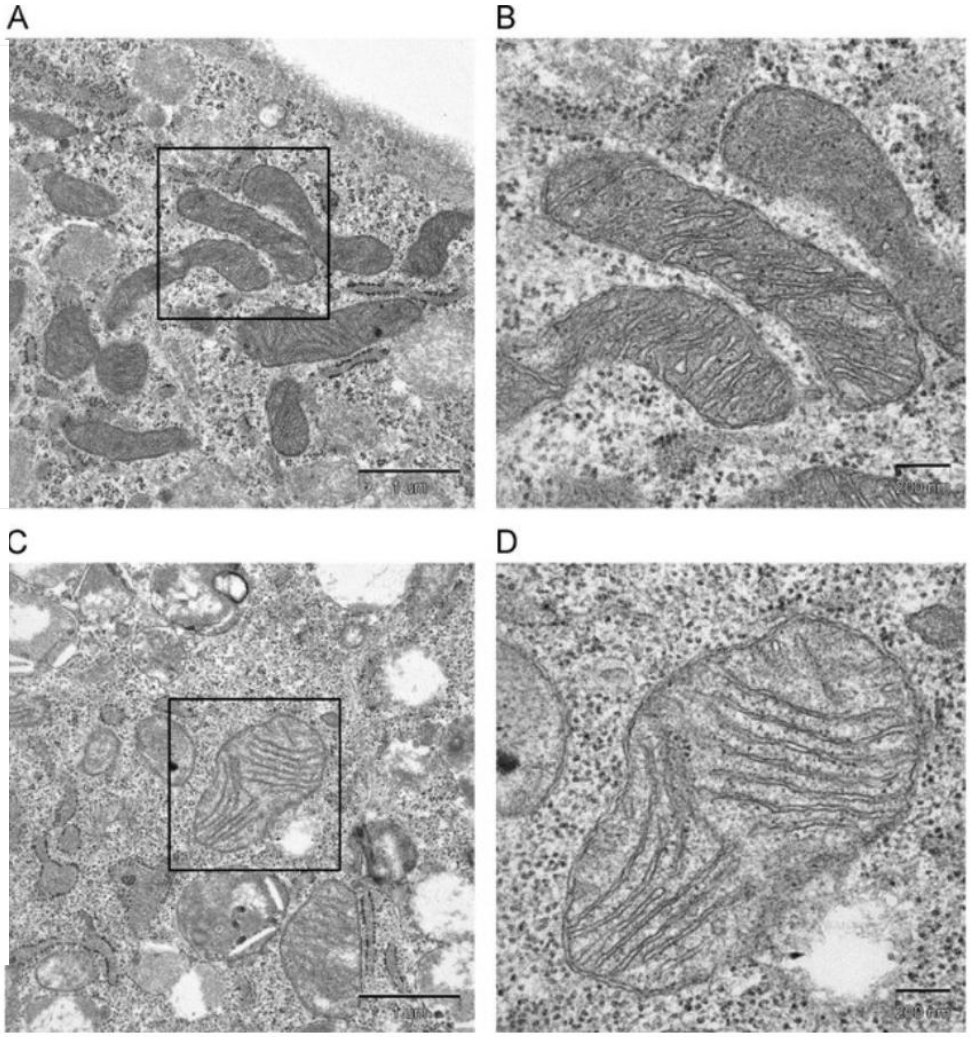


An electron micrograph of a nucleus



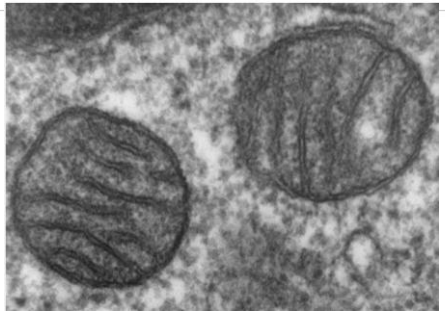




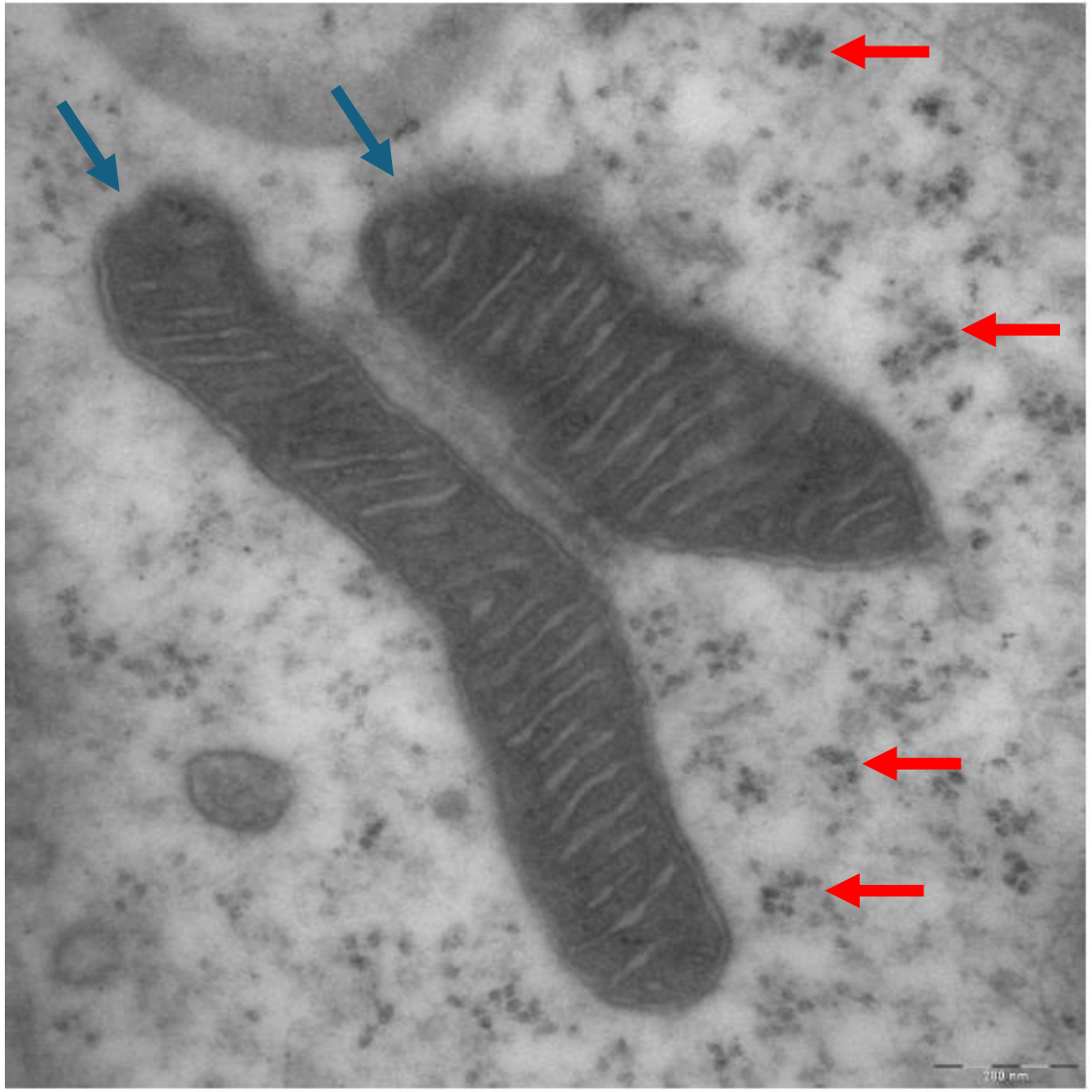


mitocondria

- Doble membrana
- Cristae
- Tamaño variable
- Alargadas o redondas



Mitocondrias y polirribosomas



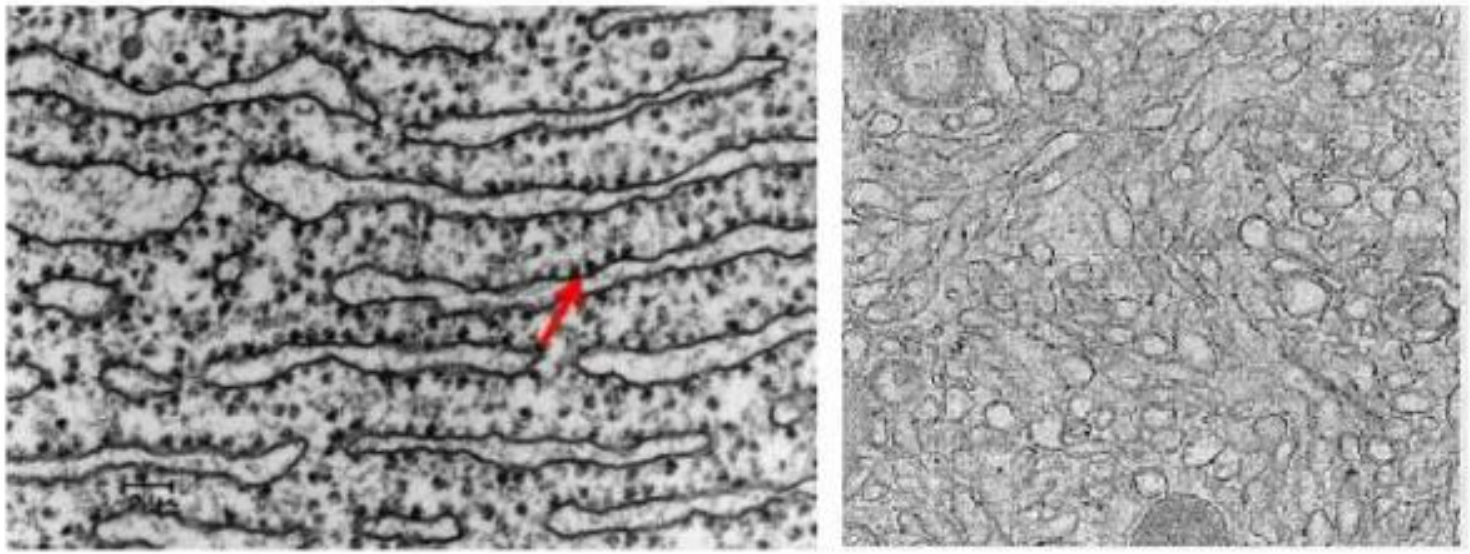
Retículo endoplasmático: rugoso vs liso; ribosomas

RE rugoso
(con ribosomas)

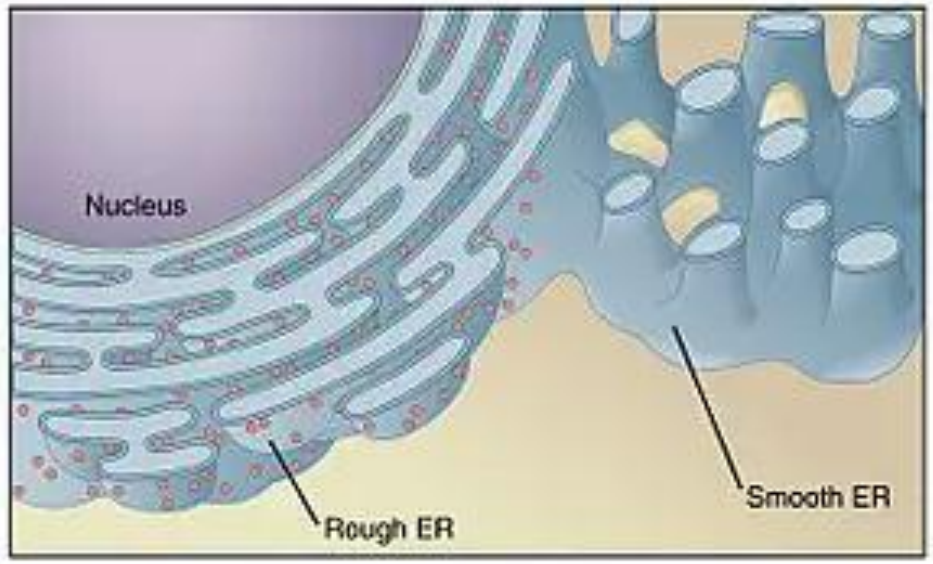
Rough endoplasmic reticulum (RER)

RE liso
(sin ribosomas)

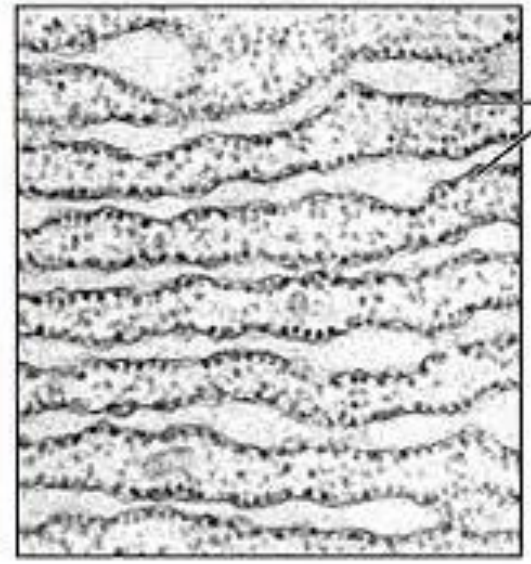
Smooth endoplasmic reticulum (SER)



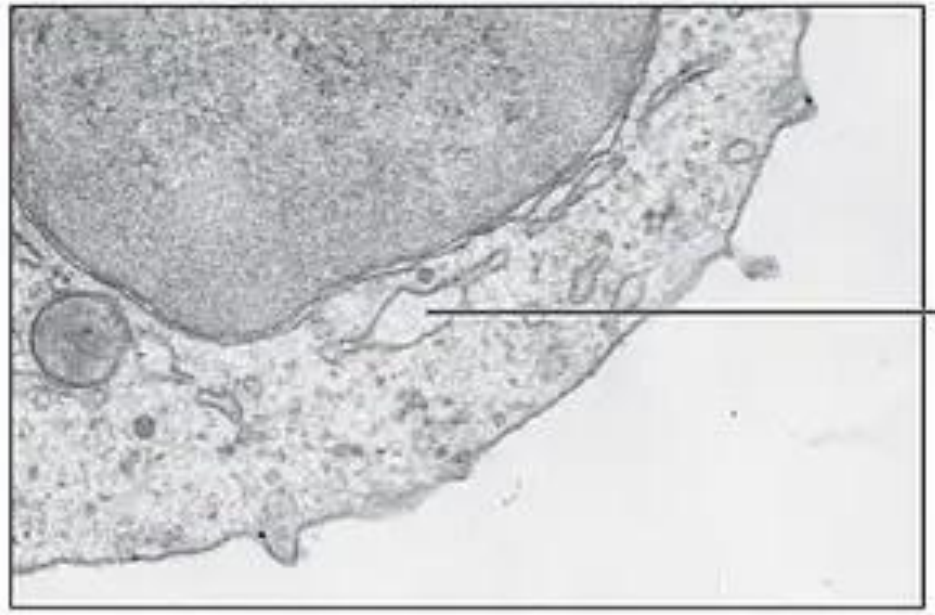
[In this figure] Left: Electron micrograph (EM) of rough ER. Red arrow indicates a ribosome particle on the cytoplasmic surface of rough ER. Right: EM of smooth ER. Photo source: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4700099/



(a)



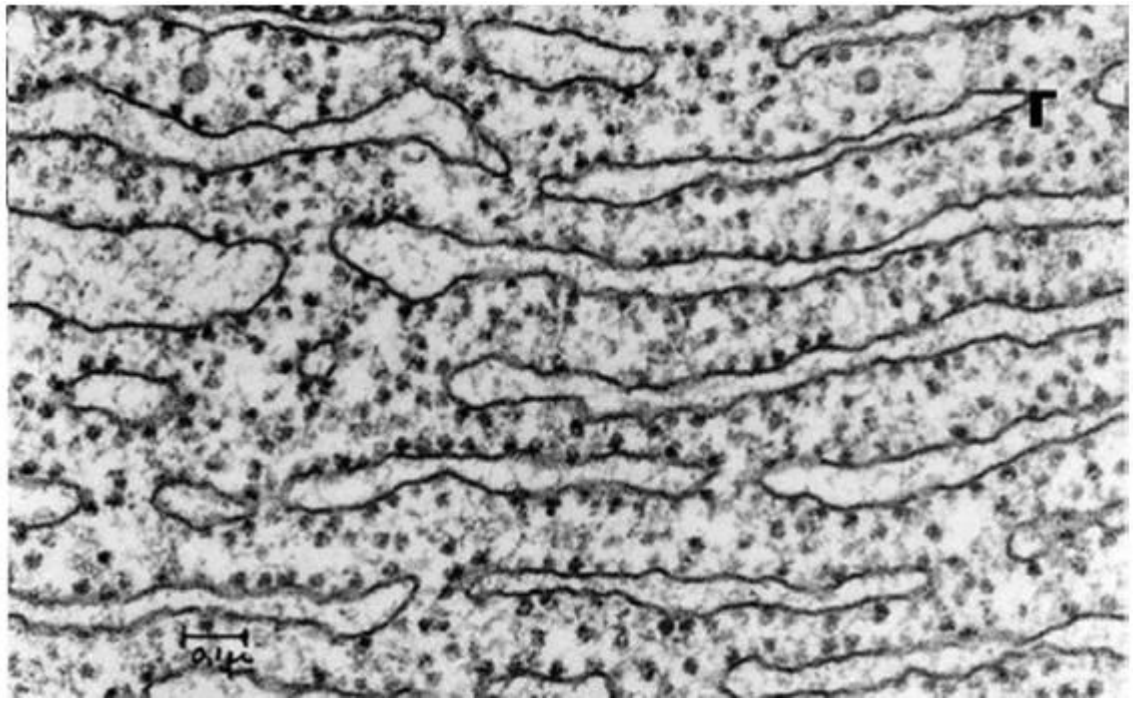
(b)

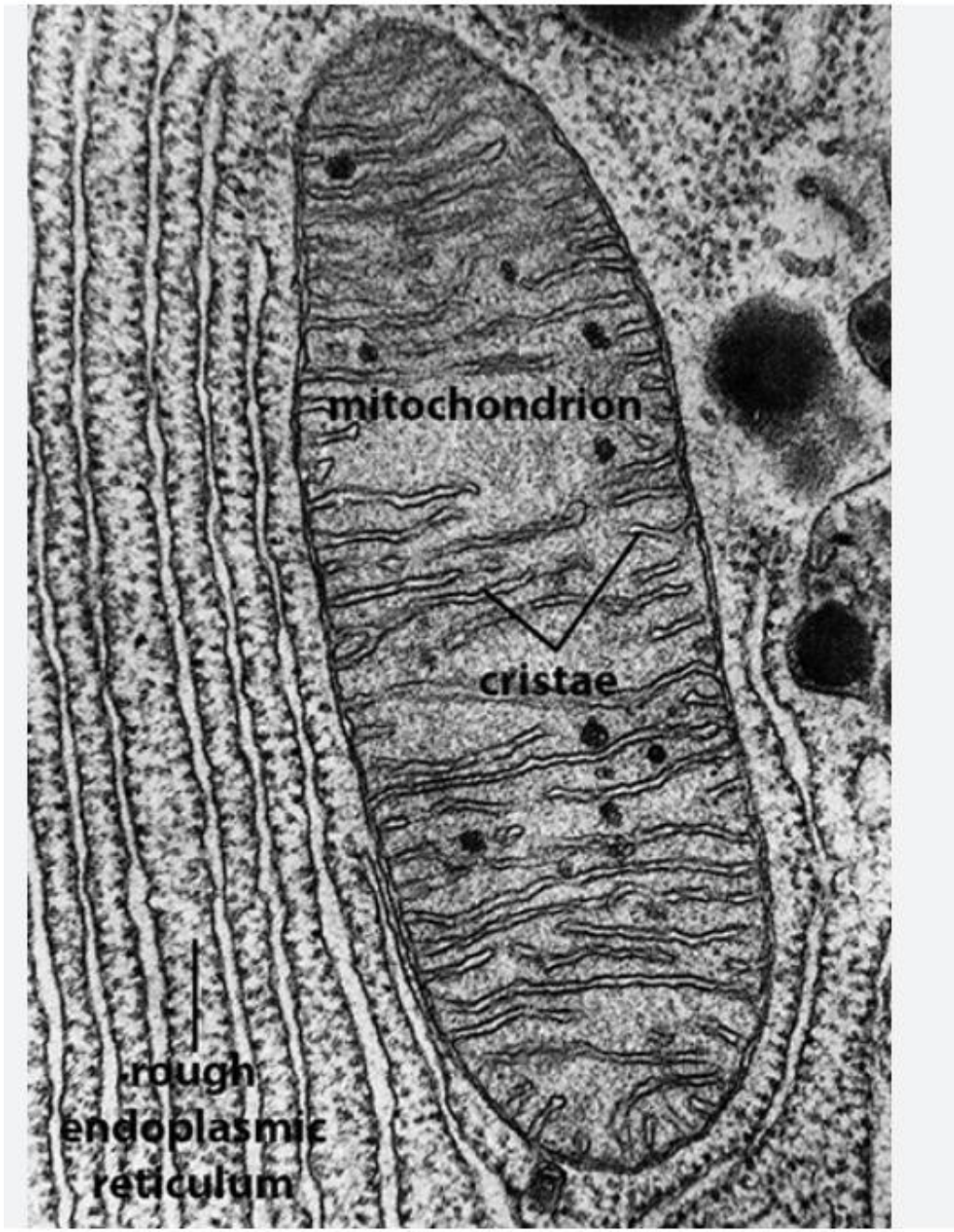


(c)

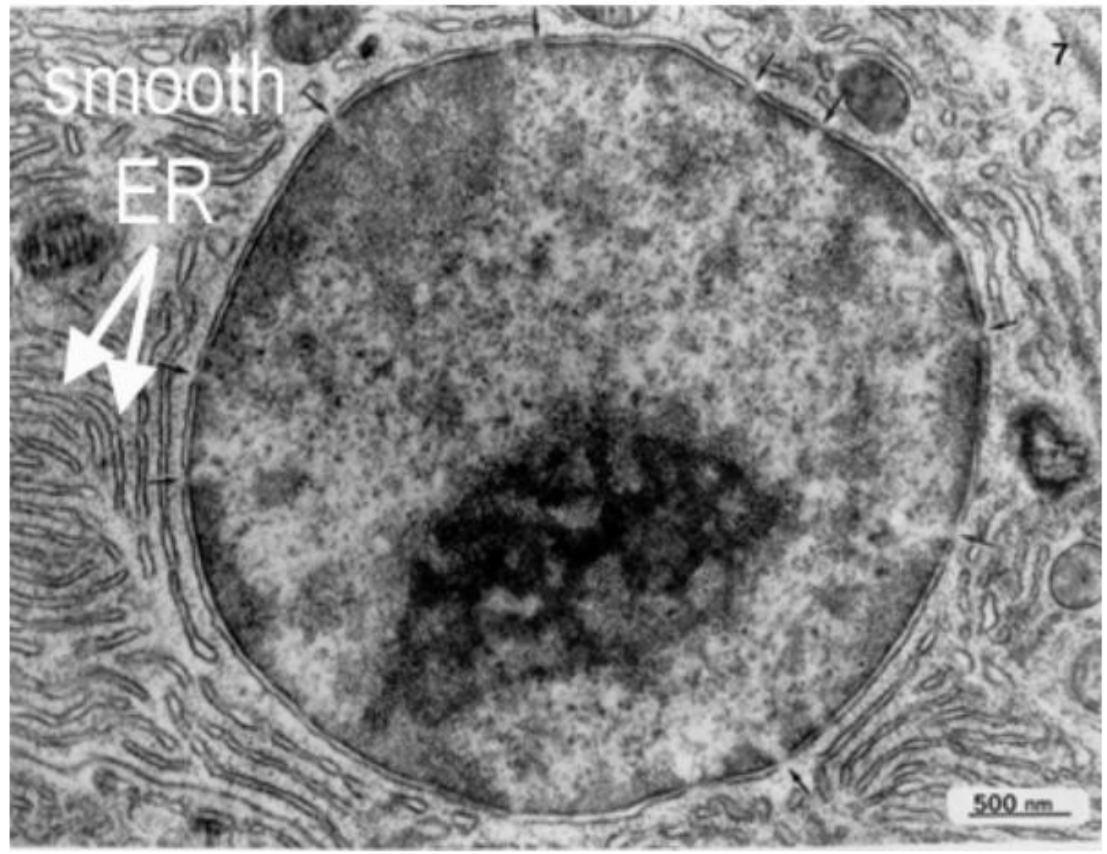
Diámetro ribosomas=15nm

RE rugoso
(con ribosomas)

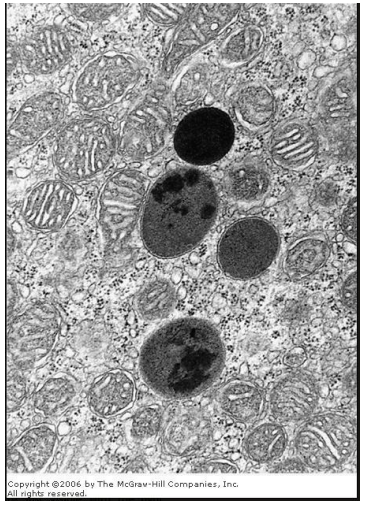




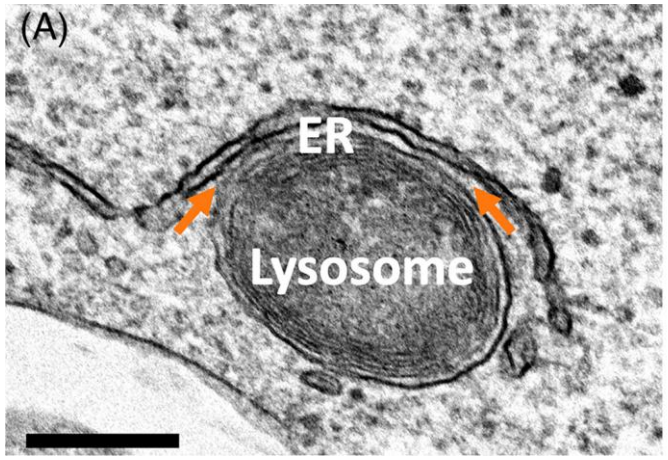
RE liso
(sin ribosomas)

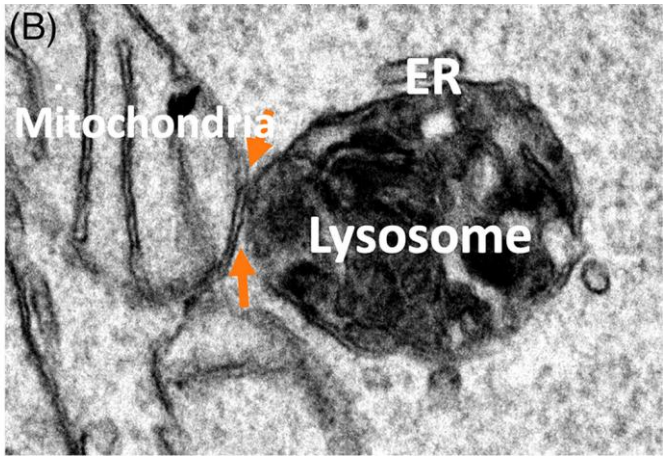


Lisosomas

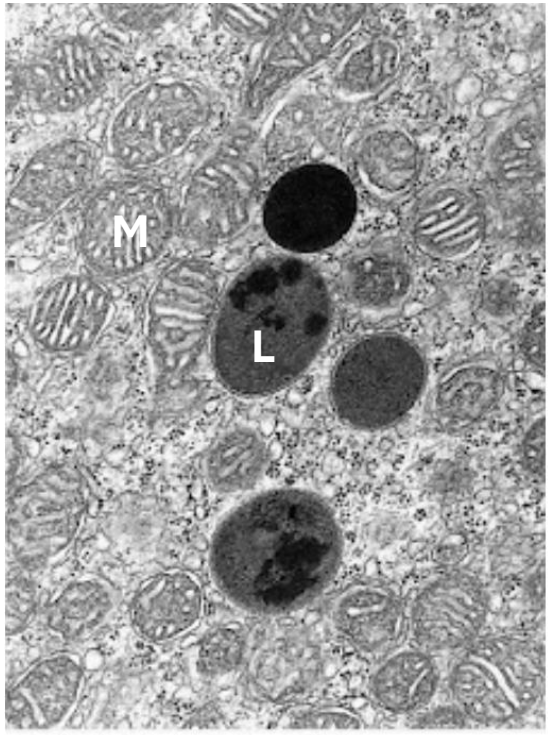






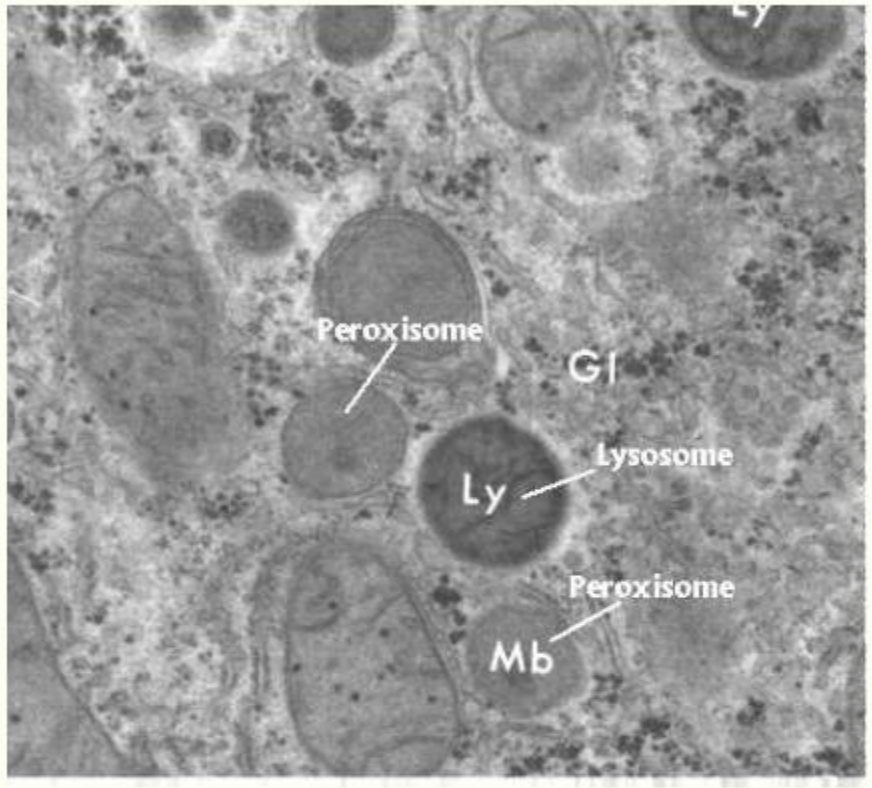


Traffic, Volume: 23, Issue: 5, Pages: 238-269, DOI: (10.1111/tra.12839)



Lisosomas vs peroxisomas

Why Peroxisomes are not like lysosomes.

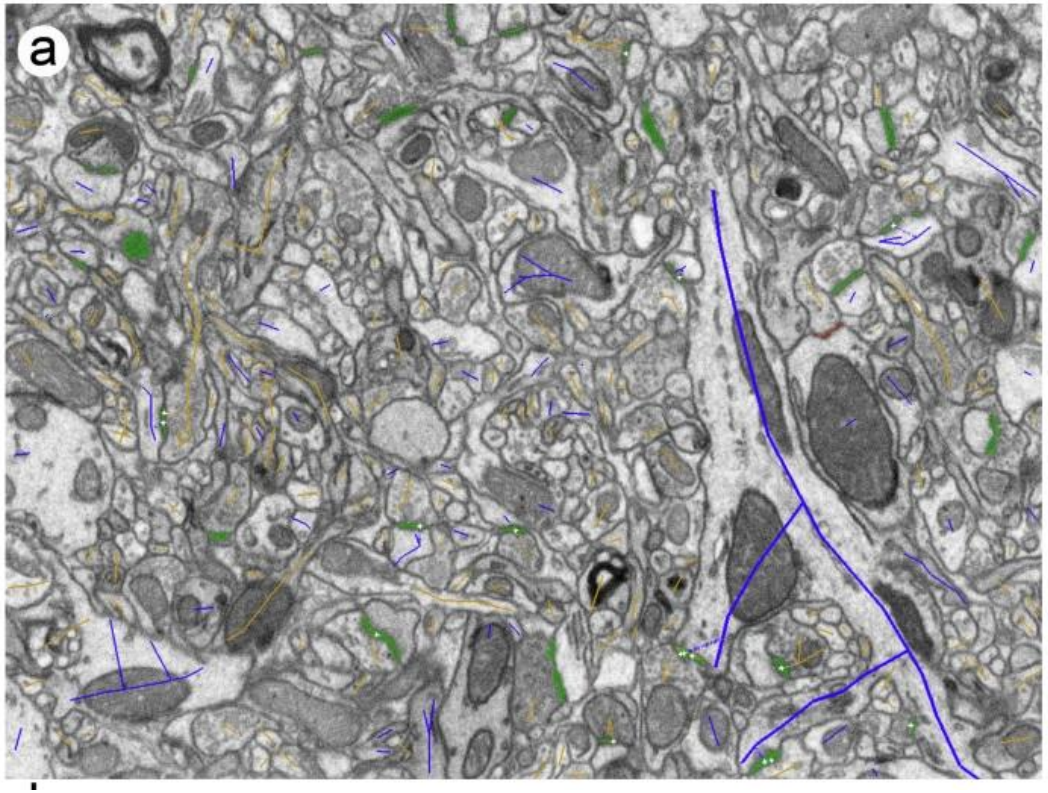


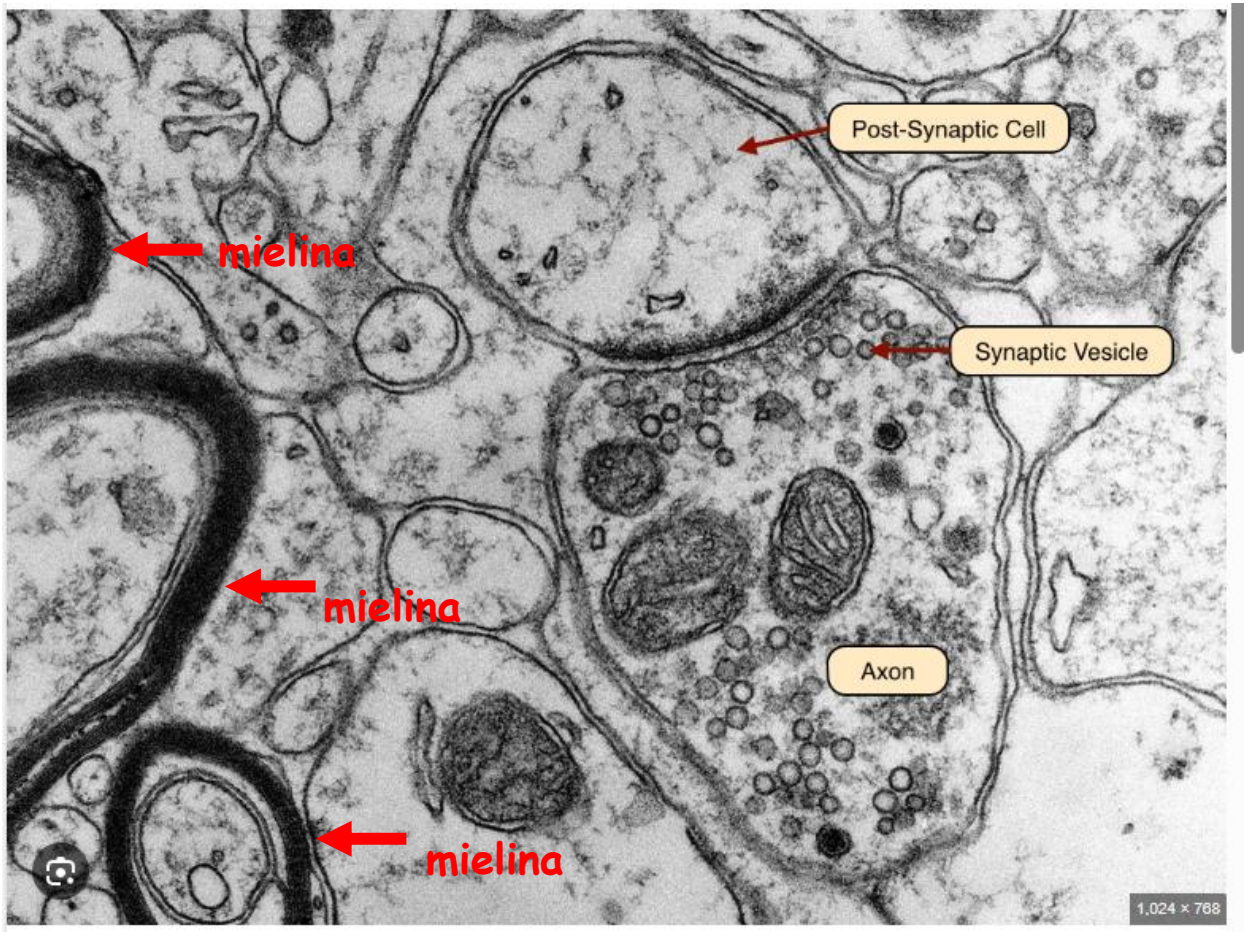
Peroxisomes are organelles that contain oxidative enzymes, such as D-amino acid oxidase, ureate oxidase, and catalase. They may resemble a lysosome, however, they are not formed in the Golgi complex. Peroxisomes are distinguished by a crystalline structure inside a sac which also contains amorphous gray material. They are self replicating, like the mitochondria. Components accumulate at a given site and they can be assembled into a peroxisome. They may look like storage granules, however, they are not formed in the same way as storage granules. They also enlarge and bud to produce new peroxisomes.

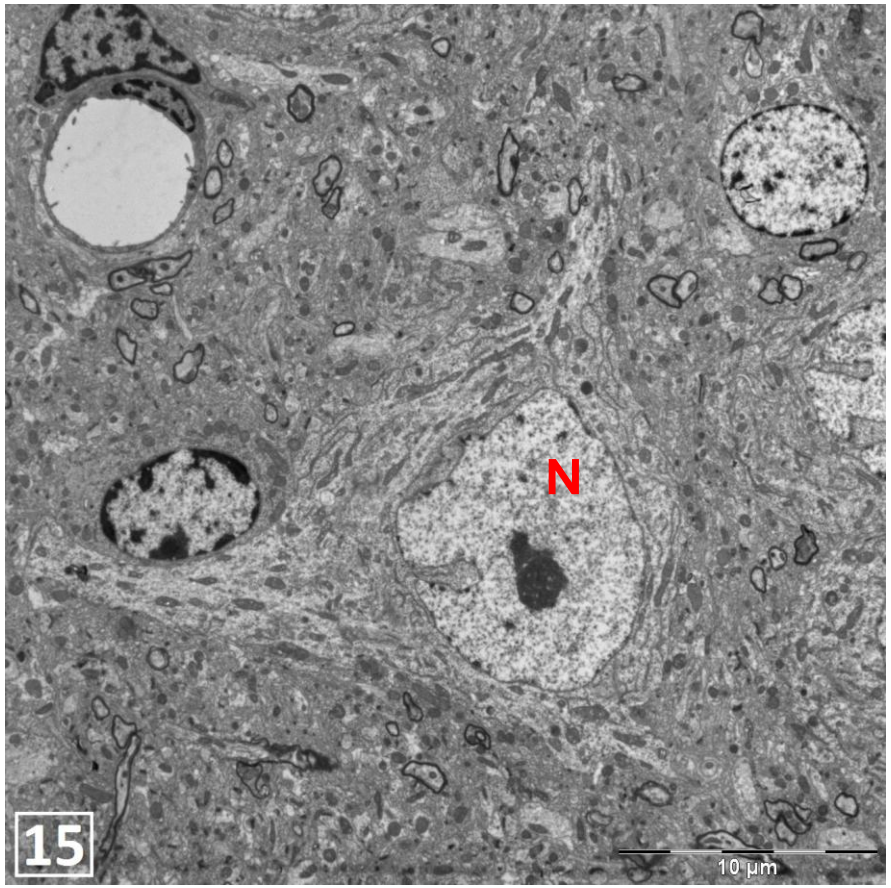
Peroxisomes function to rid the body of toxic substances like hydrogen peroxide, or other metabolites. They are a major site of oxygen utilization and are numerous in the liver where toxic byproducts are going to accumulate.

Expansiones celulares

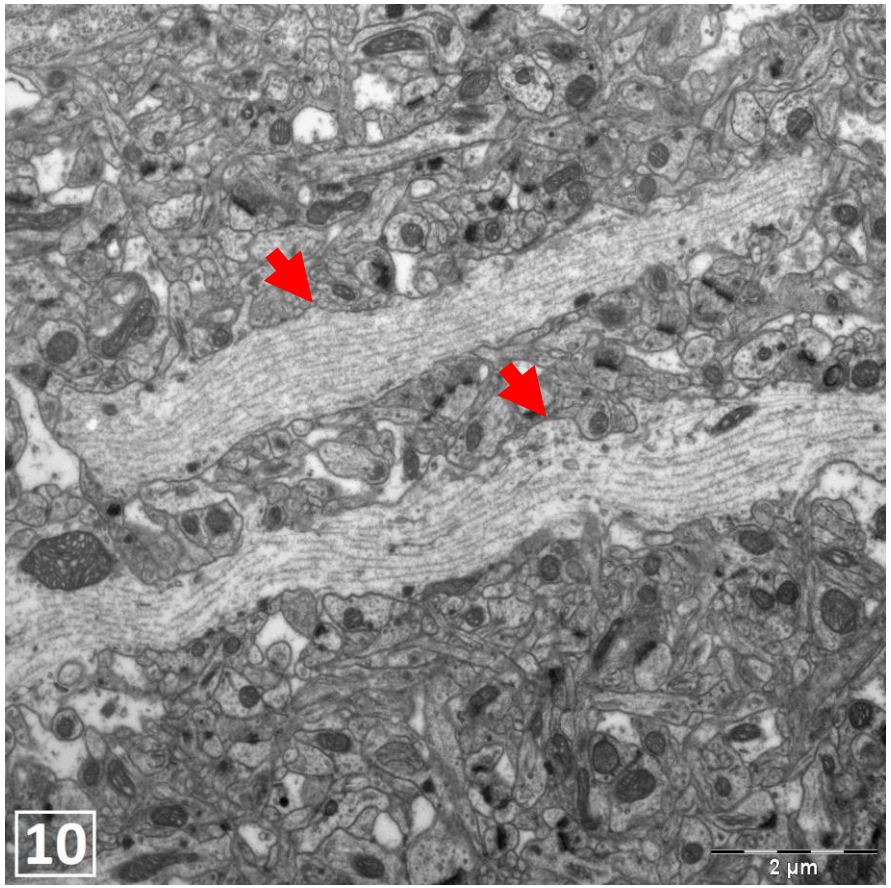
Fig. 1: Tracing dendrites and axons.





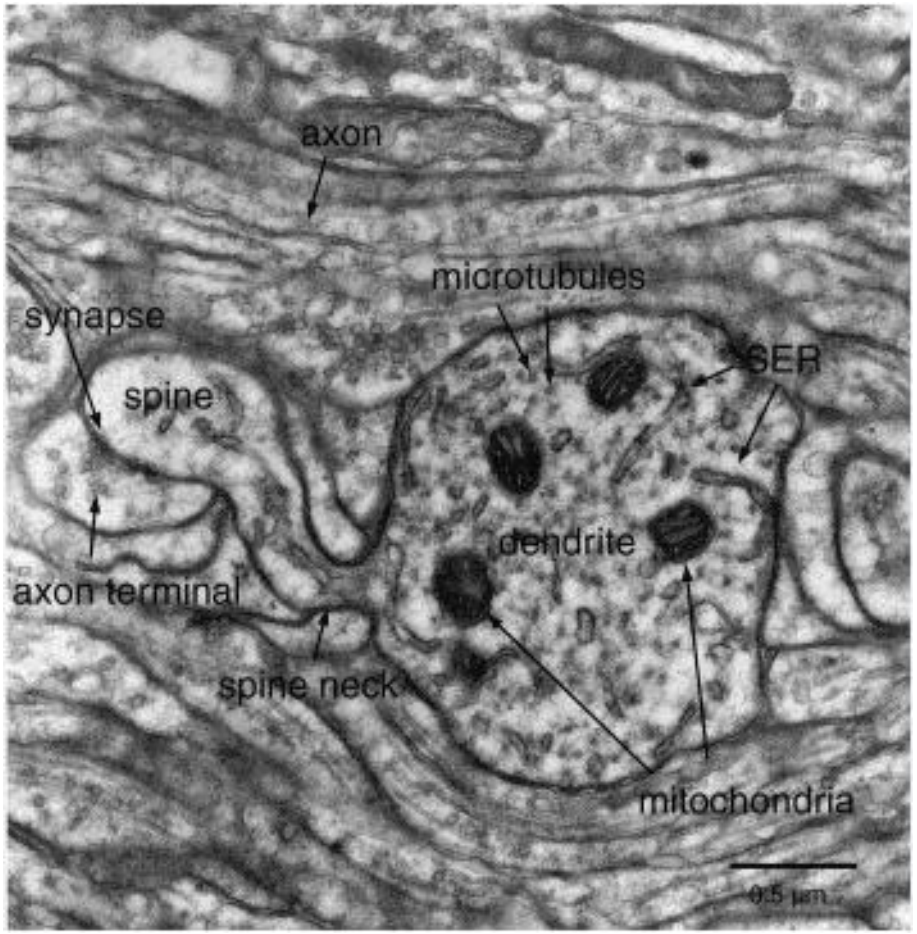


Neurona (N) y células gliales

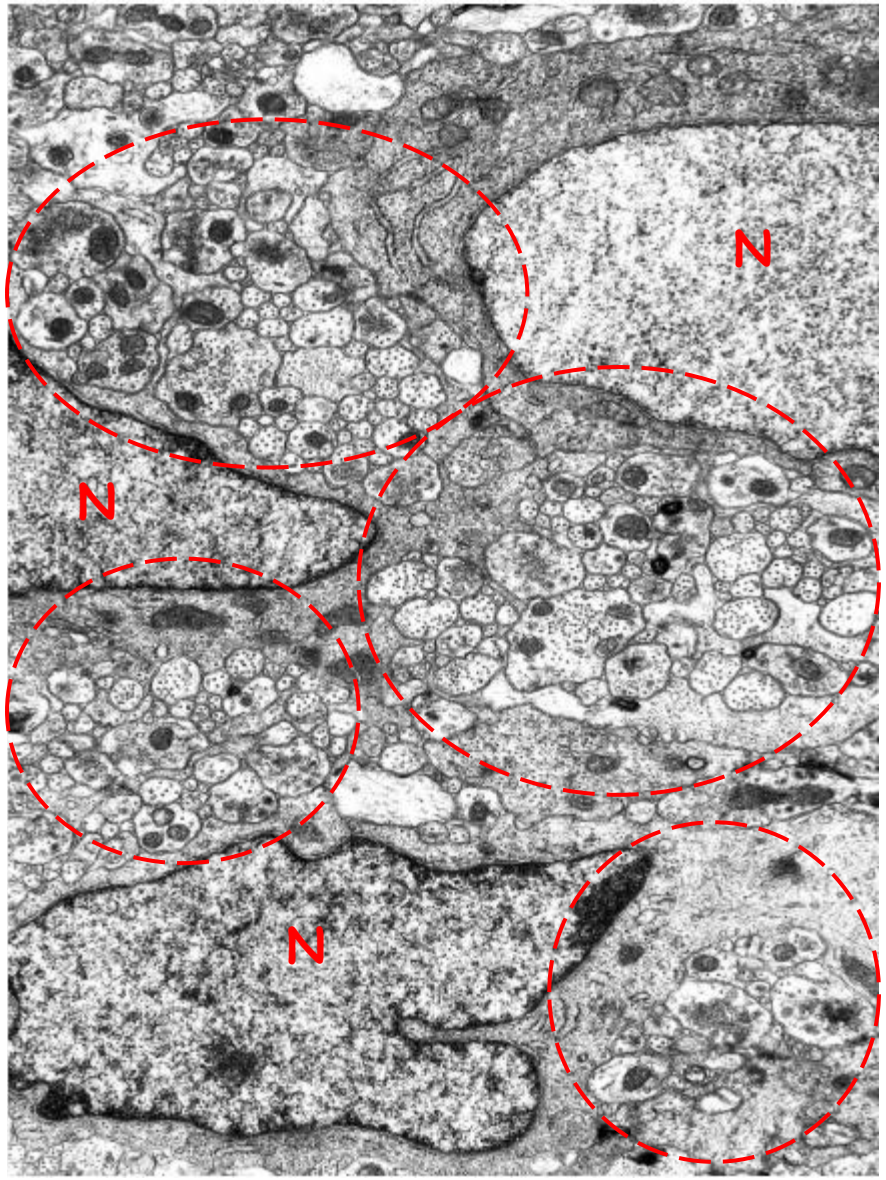


Dendritas con microtúbulos

Expansiones celulares



Expansiones celulares



Nucleos (N) y expansiones celulares (rojo)

Citoesqueleto: microtubulos vs filamentos

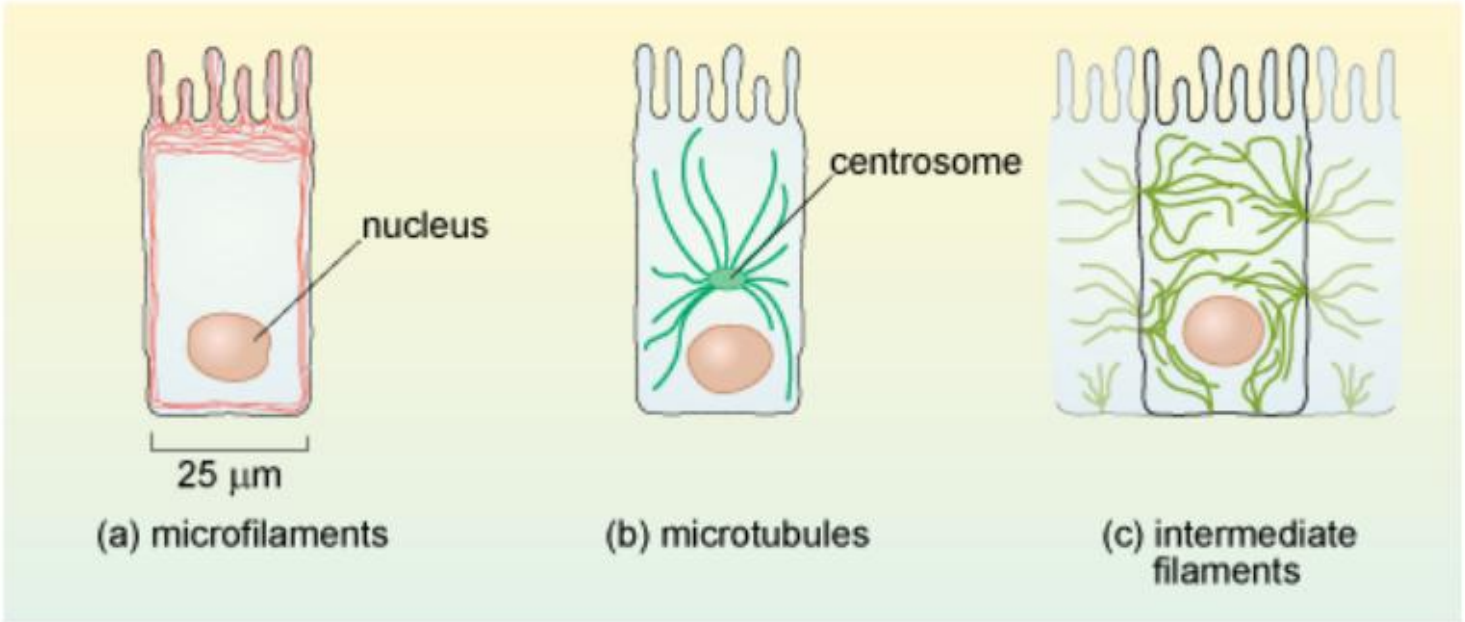


FIgure 9 The arrangement of cytoskeletal proteins in intestinal epithelial cells: (a) microfilaments form a scaffold around the perimeter of the cell; (b) microtubules radiate from the centrosome; (c) intermediate filaments link the cell to adjacent cells and the extracellular matrix at specific sites at the surface of the cell.

Microtubulos
d=25nm

- "anillo" en transversal
- "tubo" en longitudinal

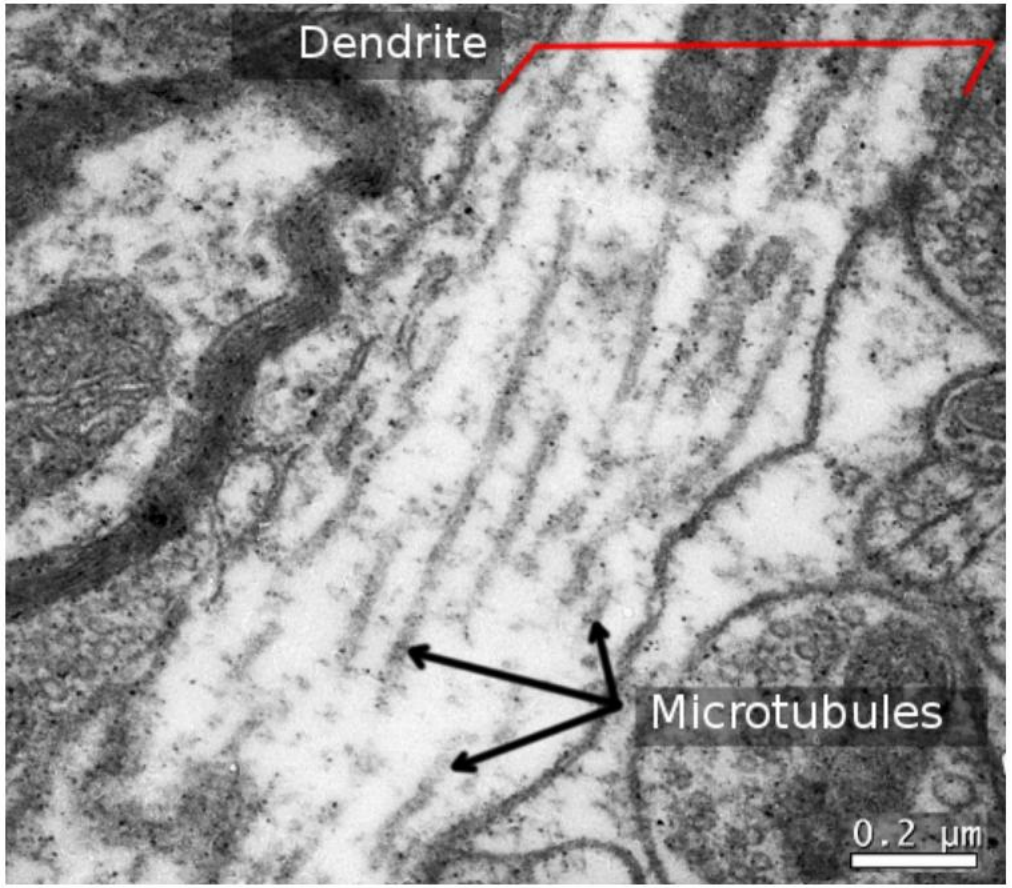
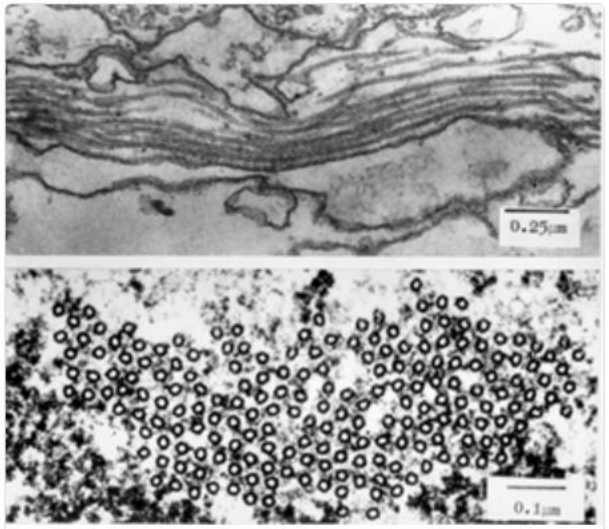


Figure 3. Transmission electron microscopy image showing microtubules within a dendrite of a neuron. Microtubules are oriented parallel to the long axis of the dendrite.

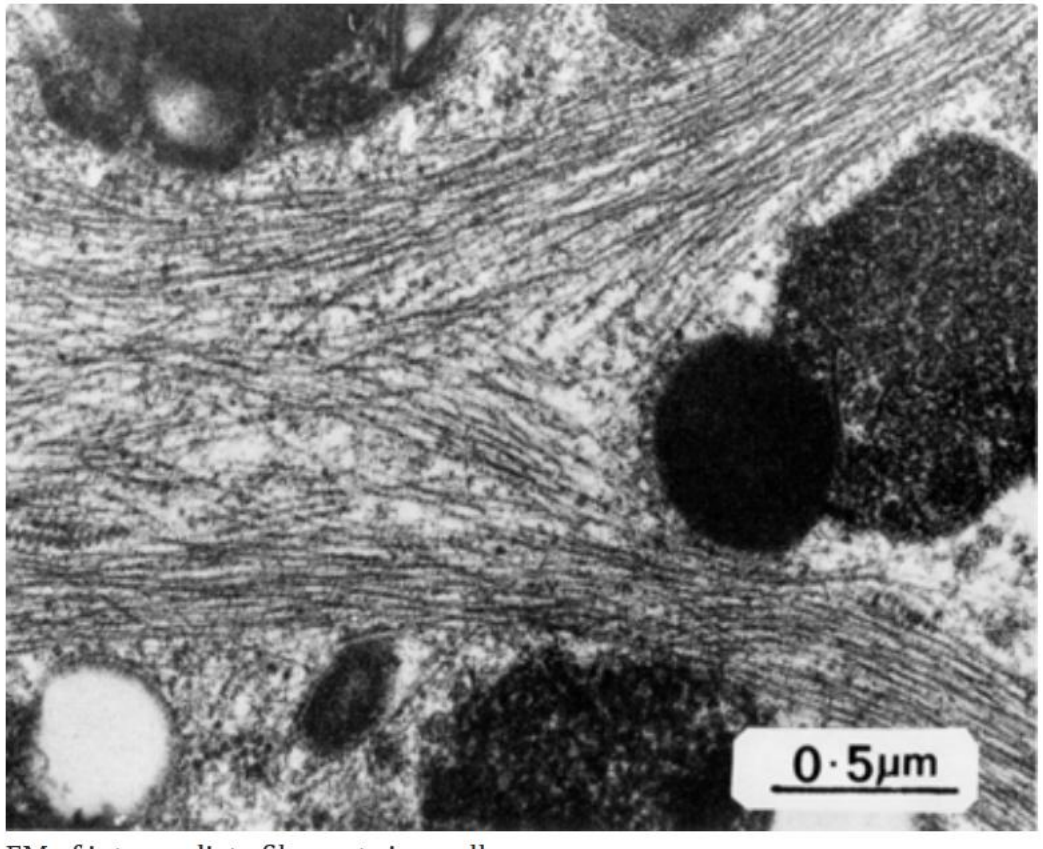


This image shows an EM of microtubules in longitudinal (top) and cross section (bottom), showing their hollow structure.

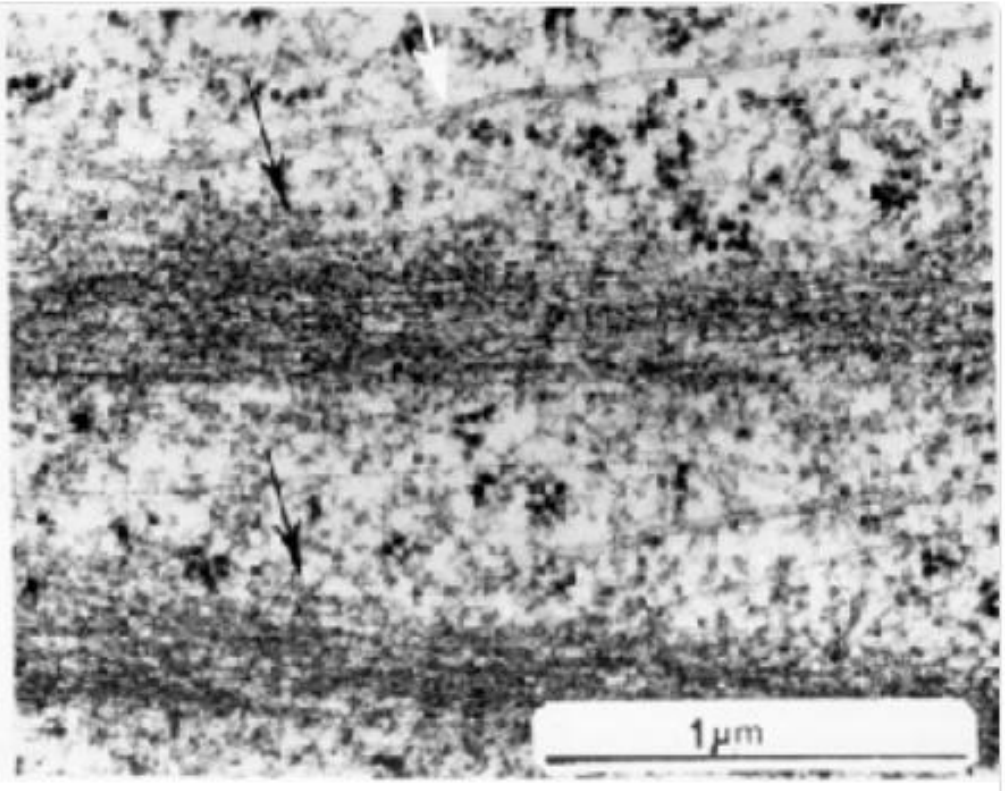
Filamentos

Intermedios d=10nm
Microfilamentos d=7nm

- Pueden ser curvos, se cruzan
- No "anillo" sino "punto"
- No "tubo" sino "hilo"



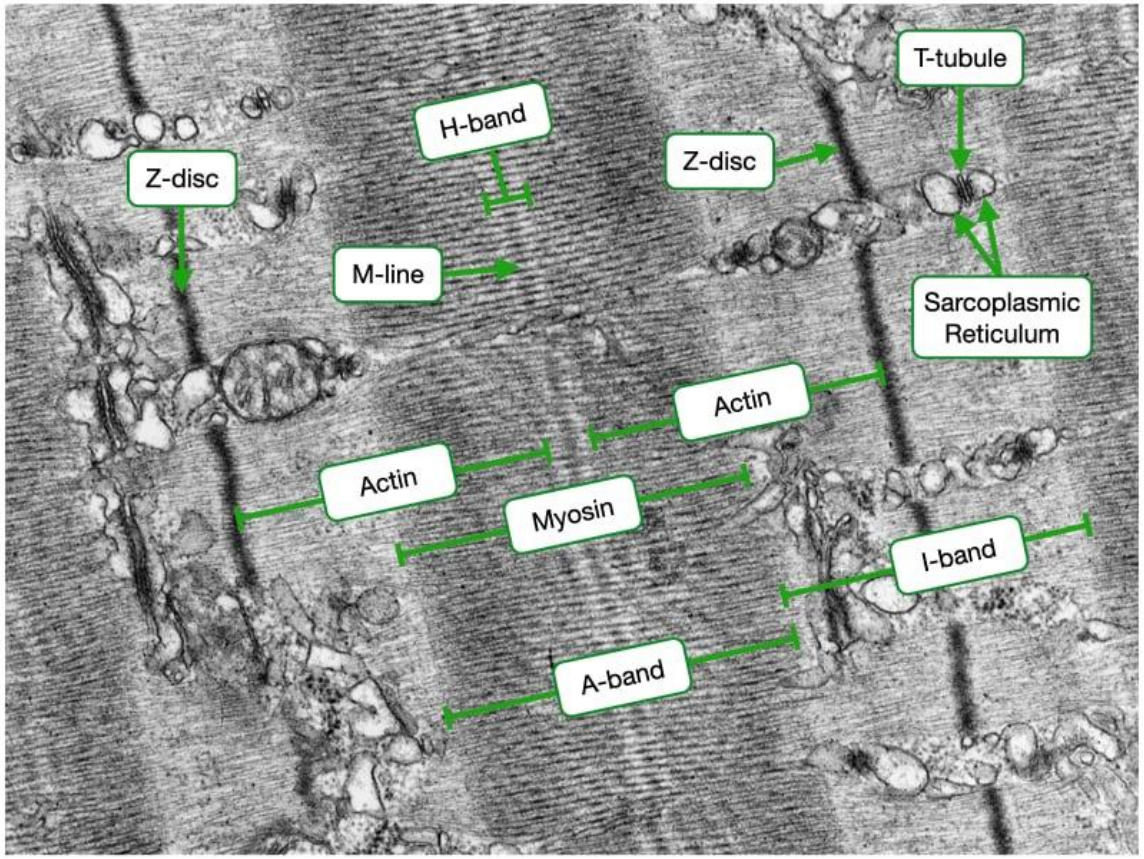
EM of intermediate filaments in a cell



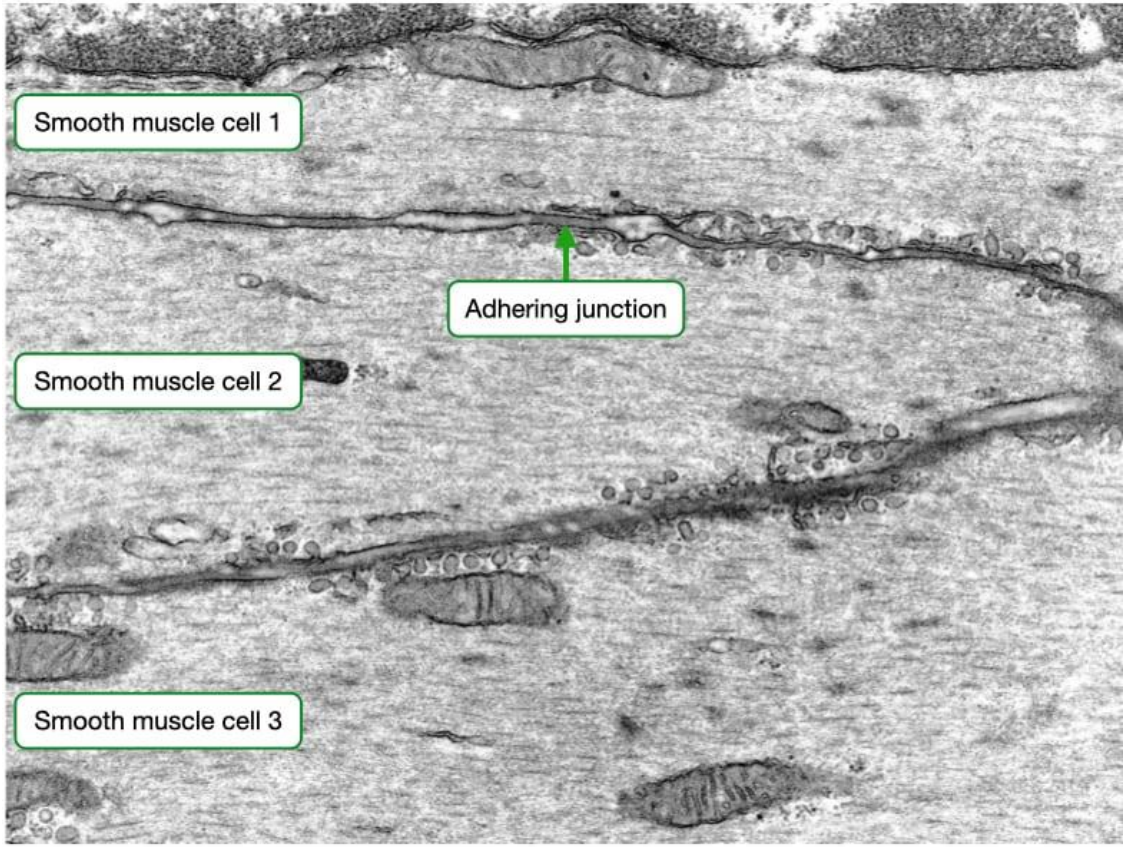
There are two bundles of microfilaments in the picture above (indicated by the black arrows)

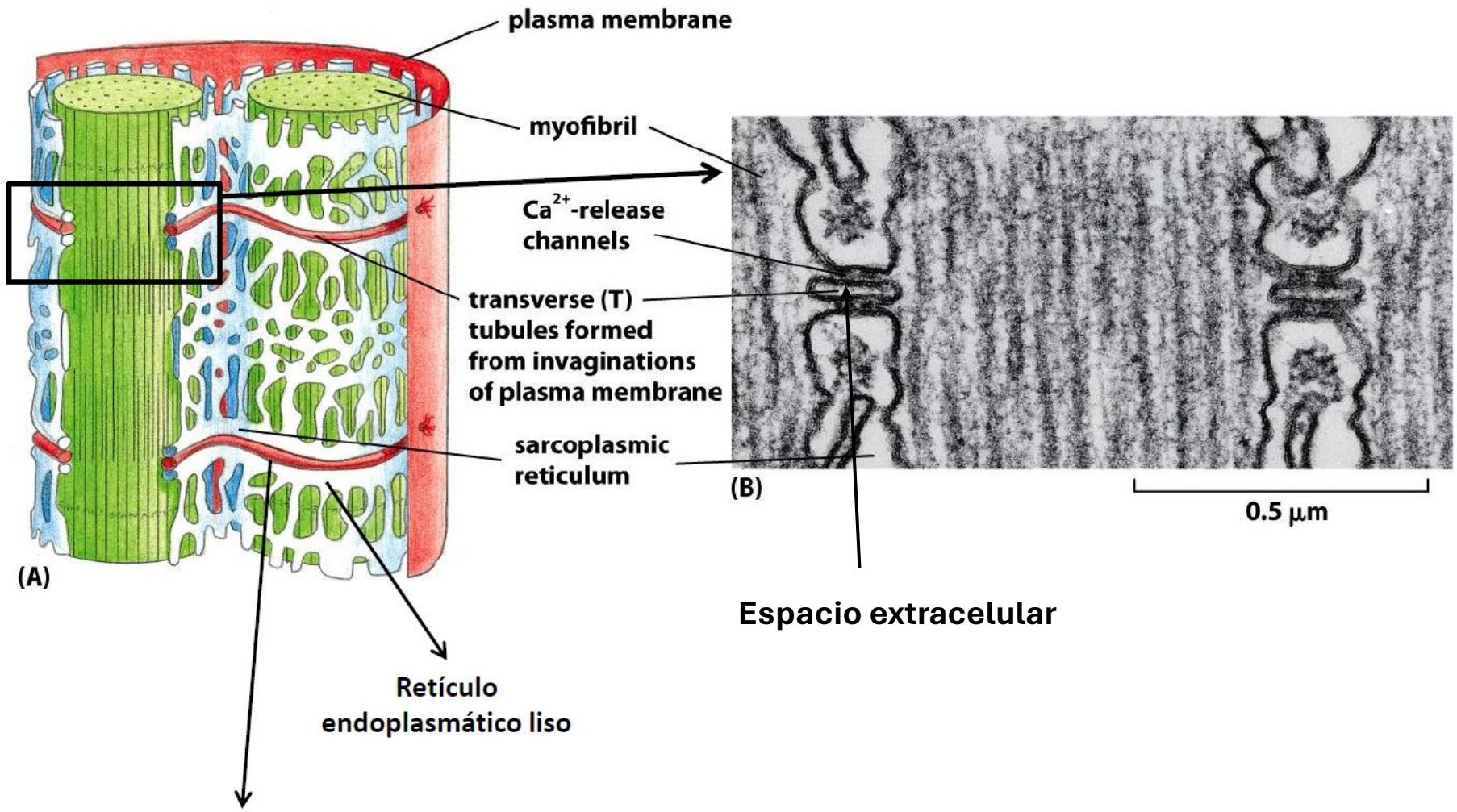
Tejido muscular

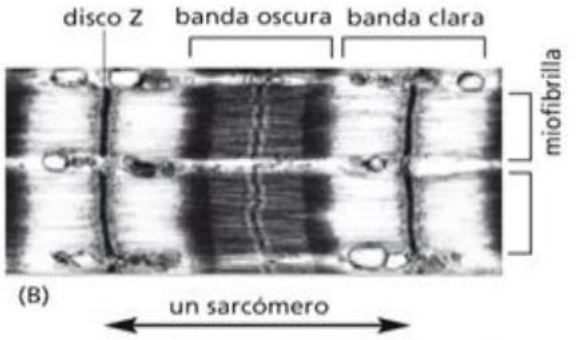
Musculo esquelético

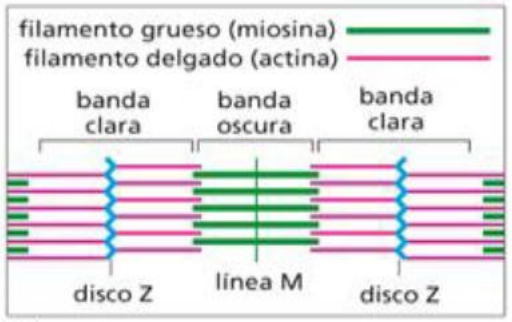


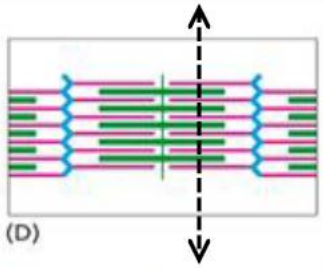
Musculo liso



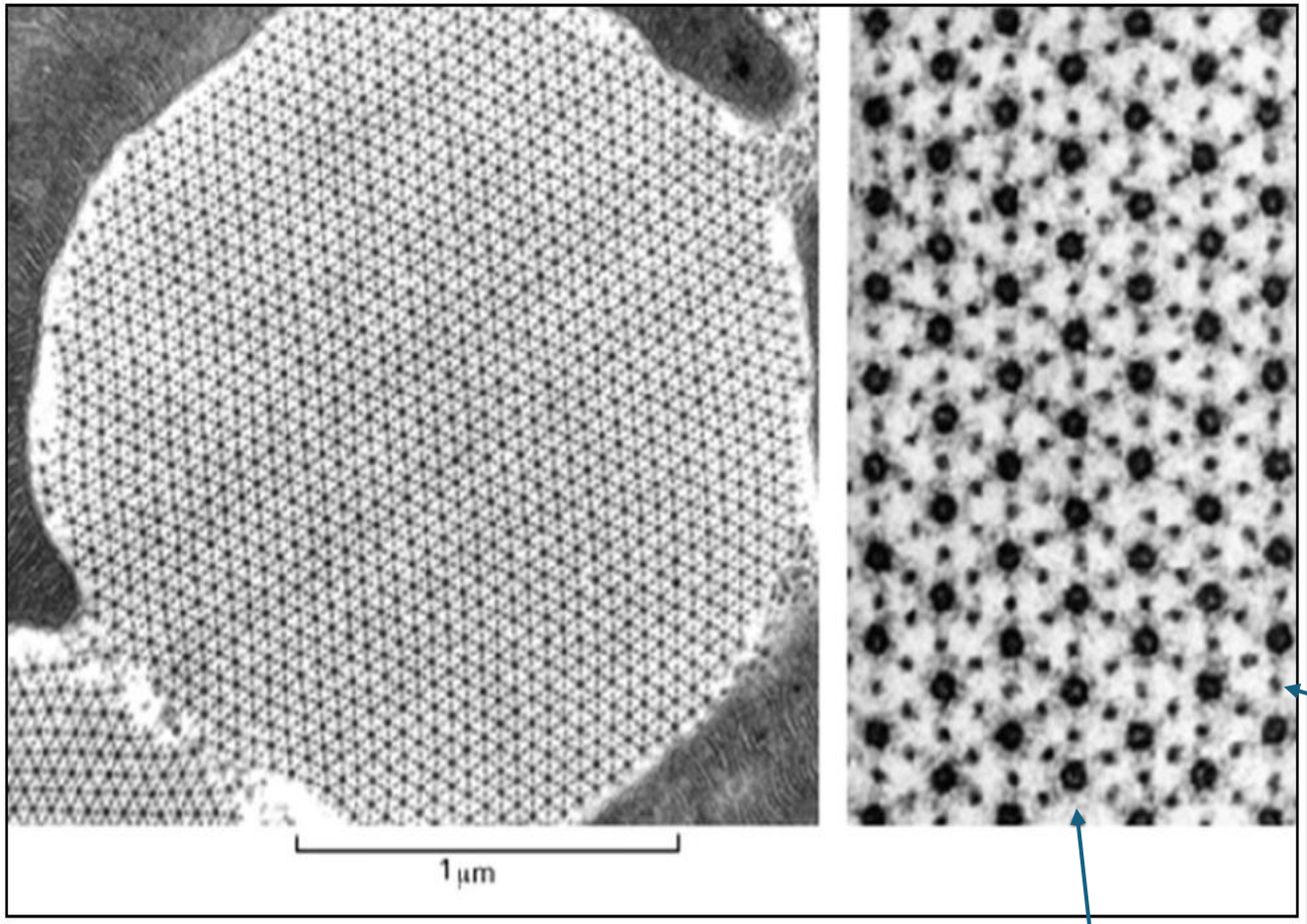

plasma membrane
myofibril
Ca^{2+}-release channels
transverse (T) tubules formed from invaginations of plasma membrane
sarcoplasmic reticulum
(A)
(B)
0.5 μm
Espacio extracelular
Retículo endoplasmático liso
Invaginaciones de la membrana plasmática de la fibra muscular: túbulos T







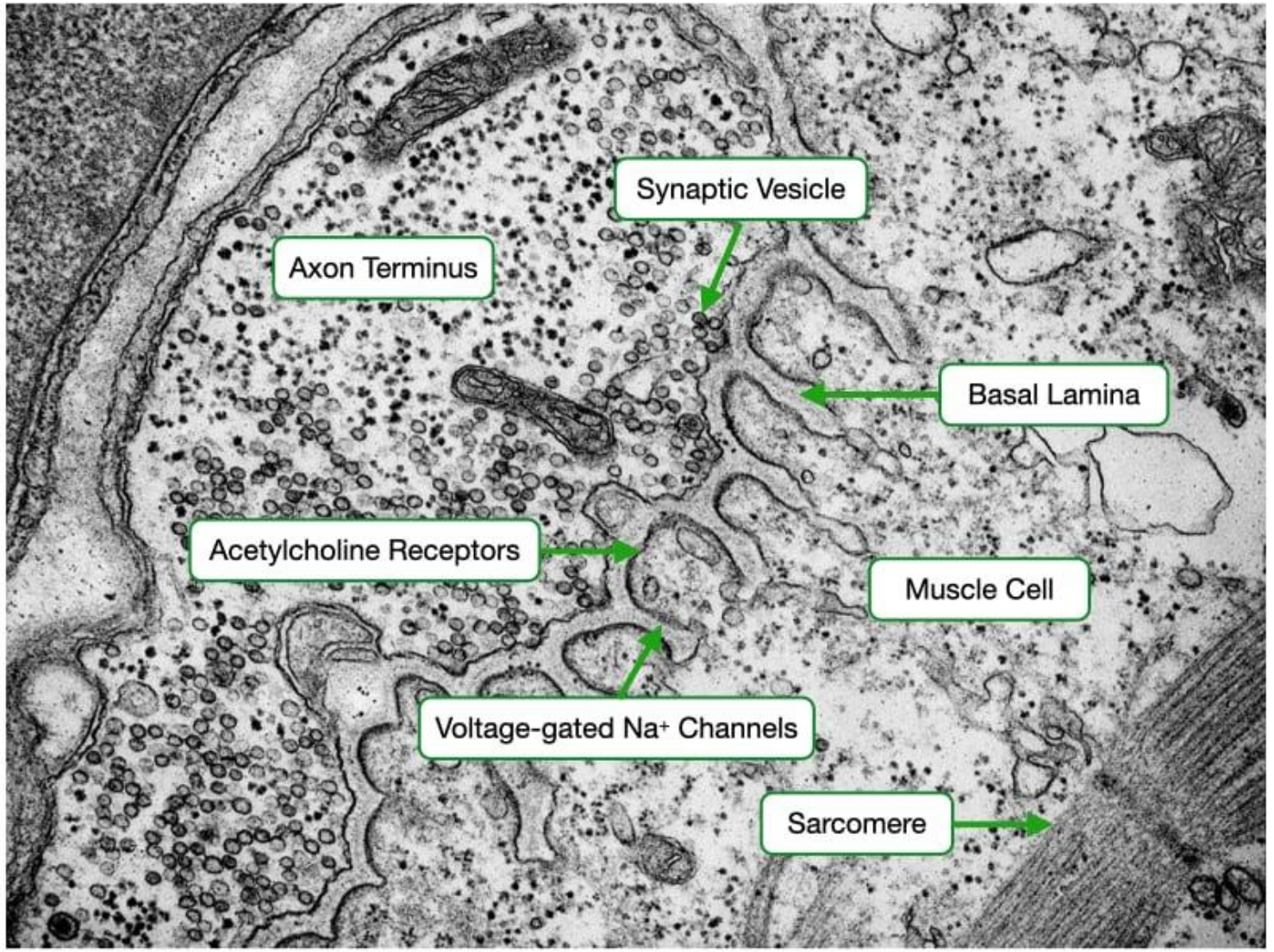
nivel del corte transversal mostrado



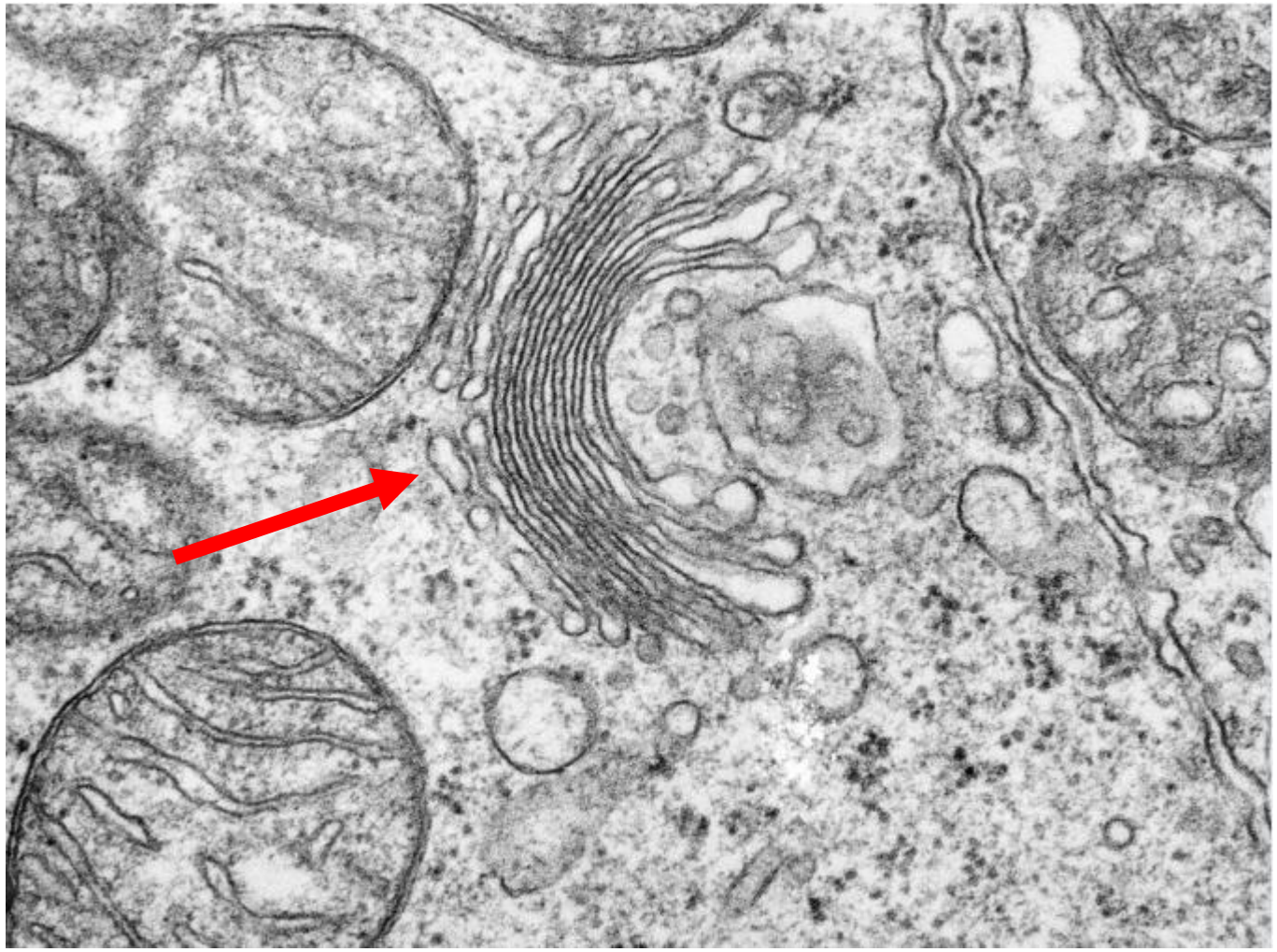
actin

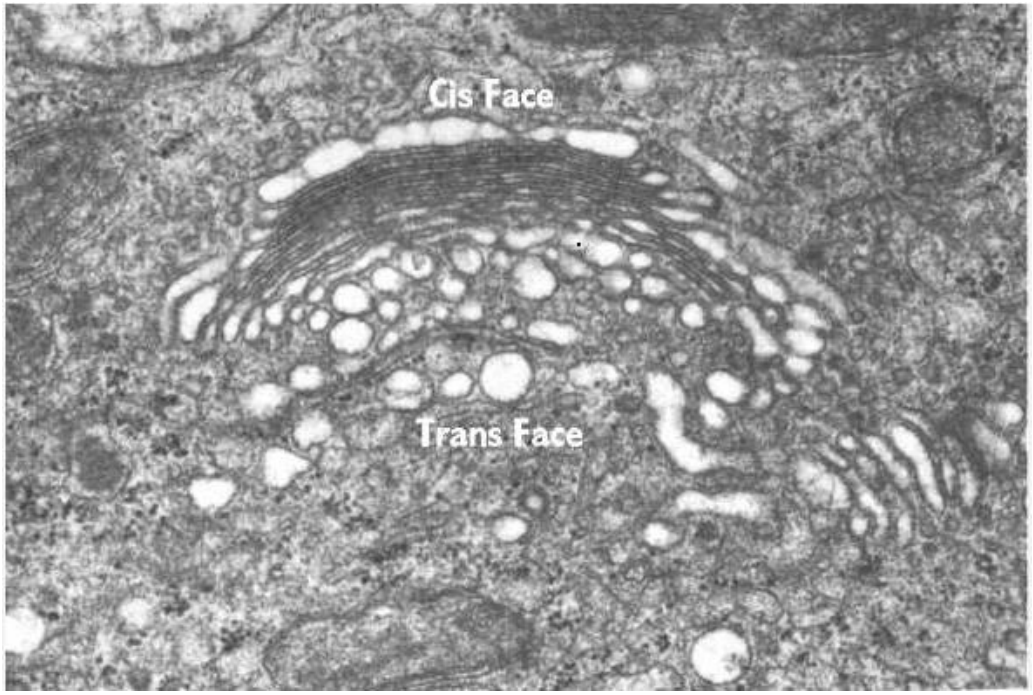
myosin

Sinapsis neuro-muscular

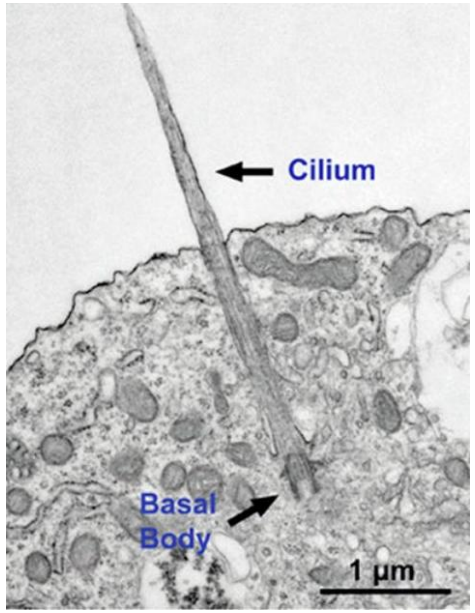


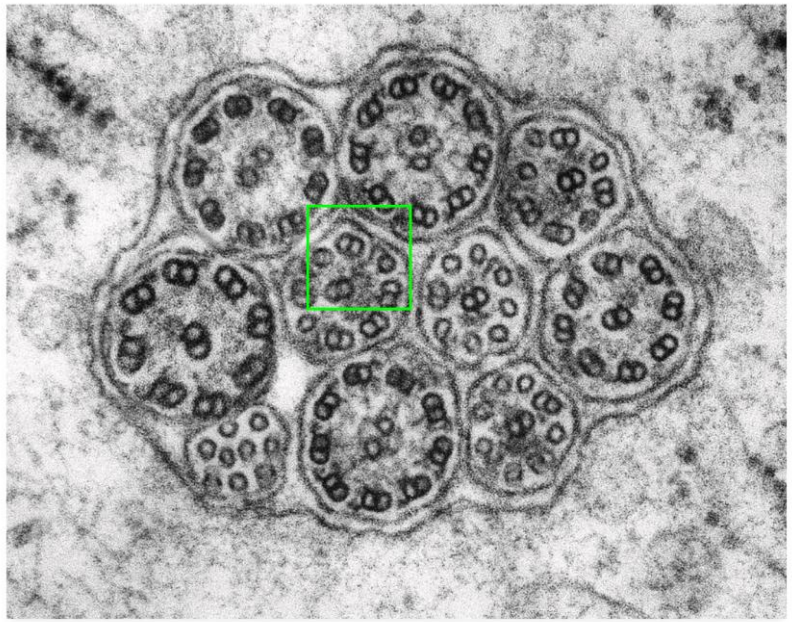
Aparato de Golgi



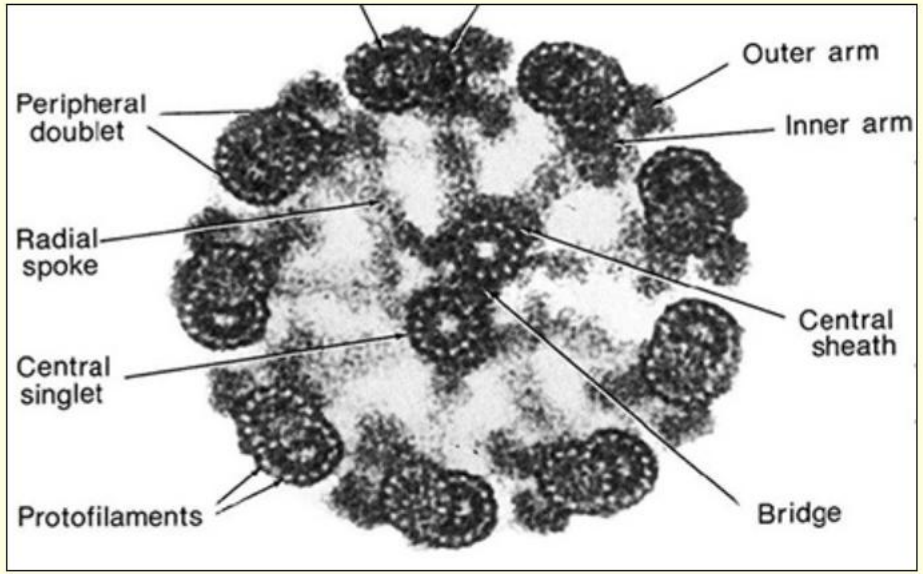


Cilio

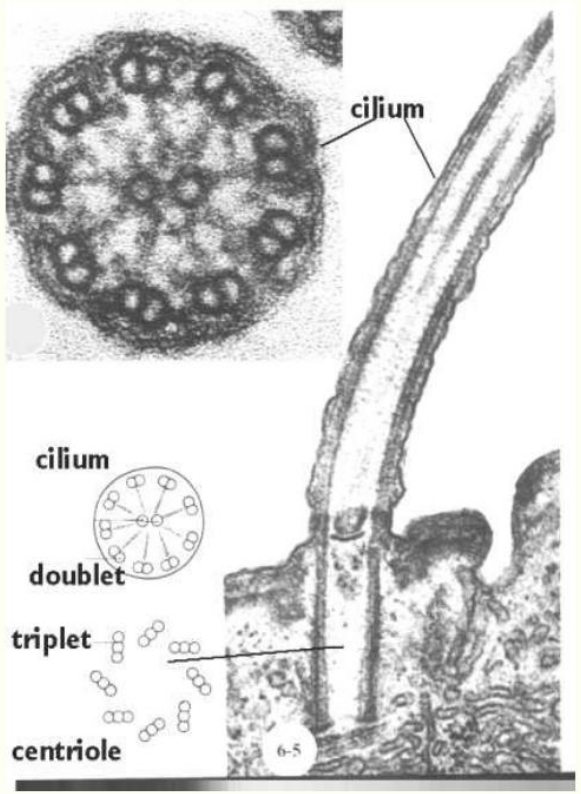




Transmission Electron Micrograph of a Cilium

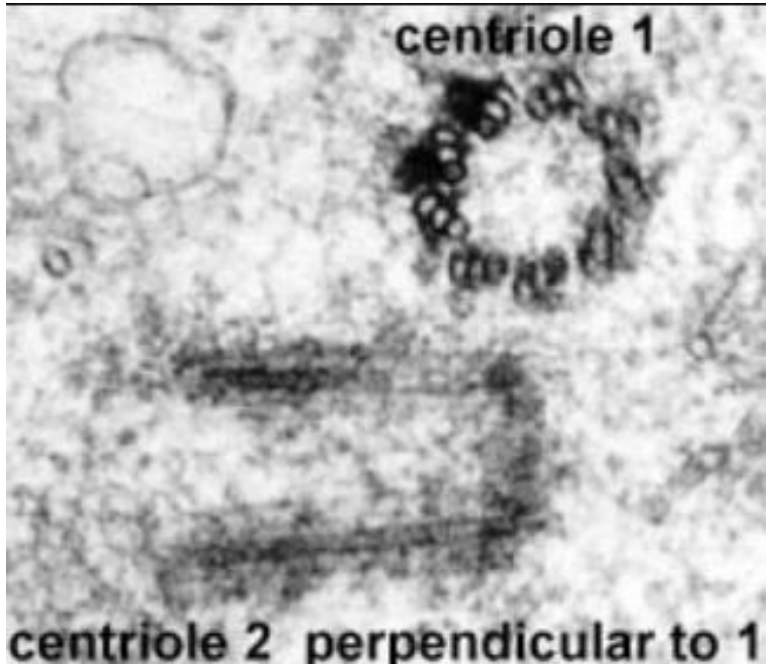


Cilia, flagella, and centrioles

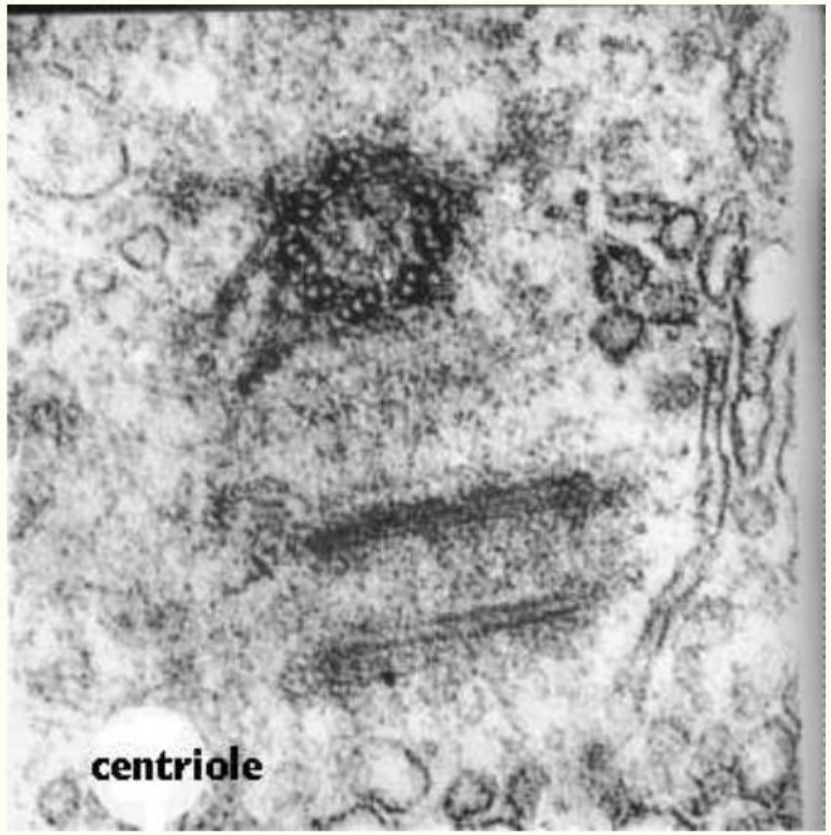


Cilia and flagella are projections from the cell. They are made up of microtubules , as shown in this cartoon and are covered by an extension of the plasma membrane. They are motile and designed either to move the cell itself or to move substances over or around the cell. The primary purpose of cilia in mammalian cells is to move fluid, mucous, or cells over their surface. Cilia and flagella have the same internal structure. The major difference is in their length.

Centriolo



Centriole structure



Like Cilia and Flagella, Centrioles are also made of microtubules. The difference is that they contain 9 sets of triplets and no doublet in the center. How the triplets in the basal body turn into the cilium doublet remains a mystery. Centrioles come in pairs, each organized at right angles to the other. This figure shows an electron micrograph of a pair of centrioles and the cartoon compares the cross section of a cilium with that of a centriole. Centrioles organize the spindle apparatus on which the chromosomes move during mitosis.

Vesículas

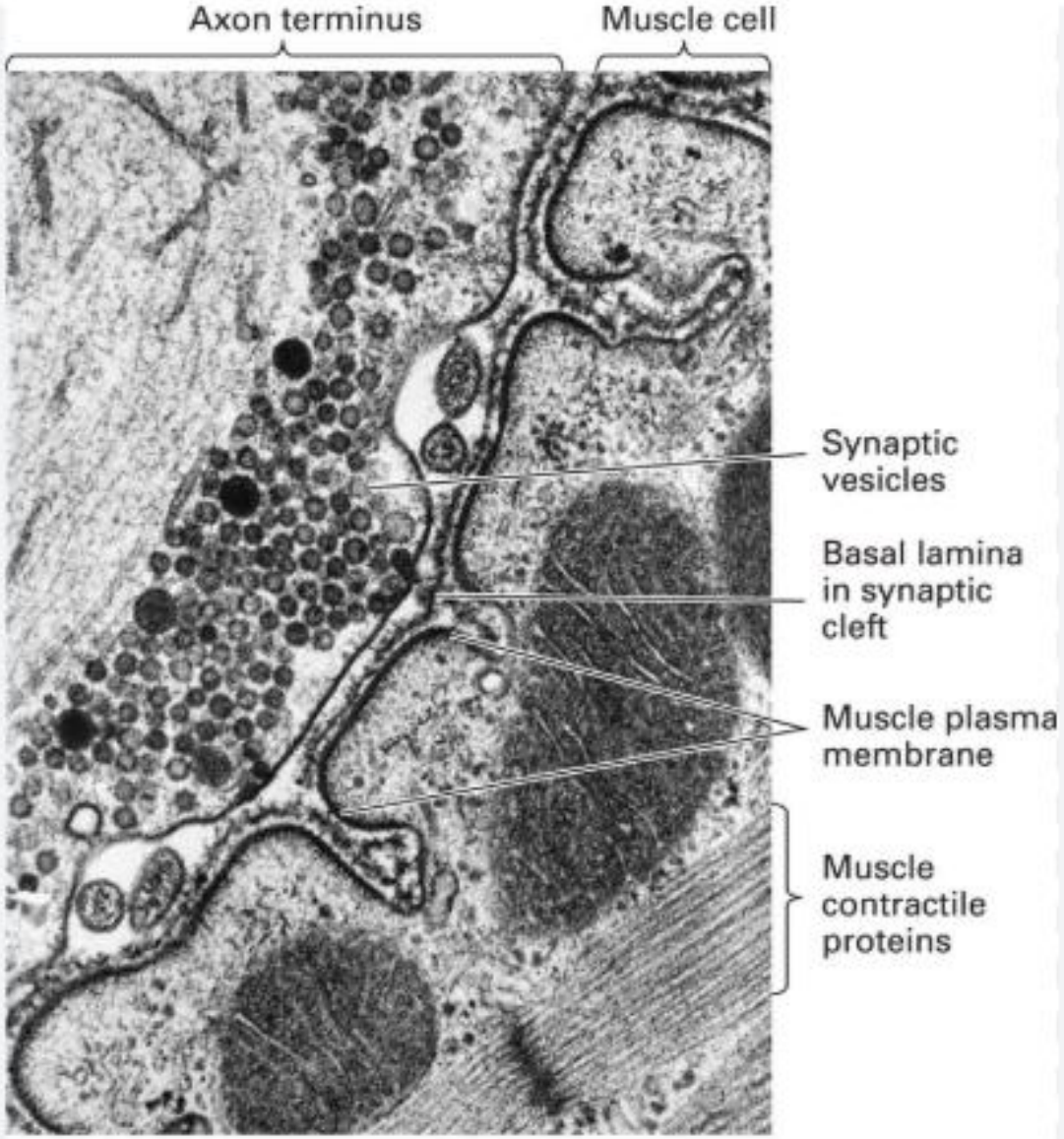
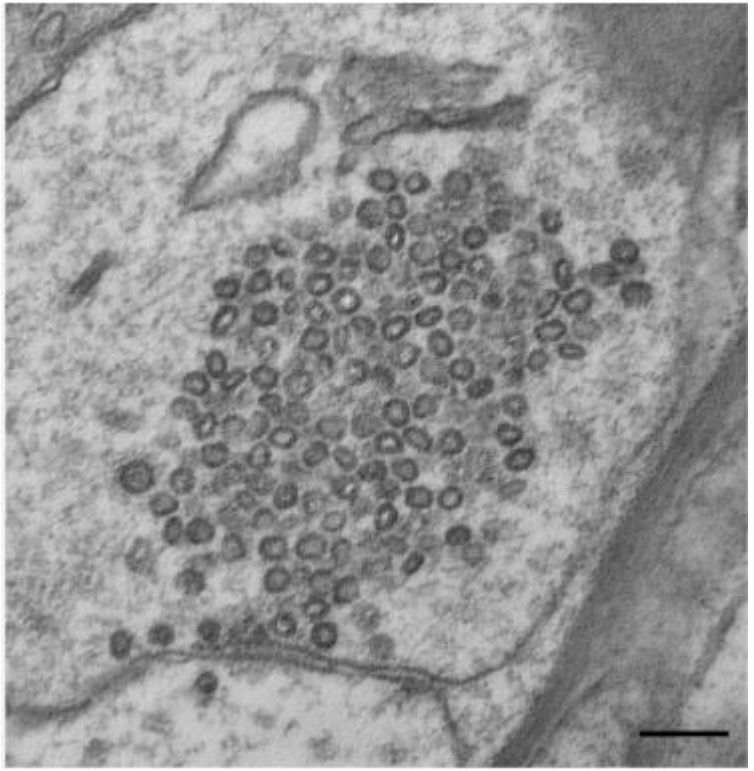
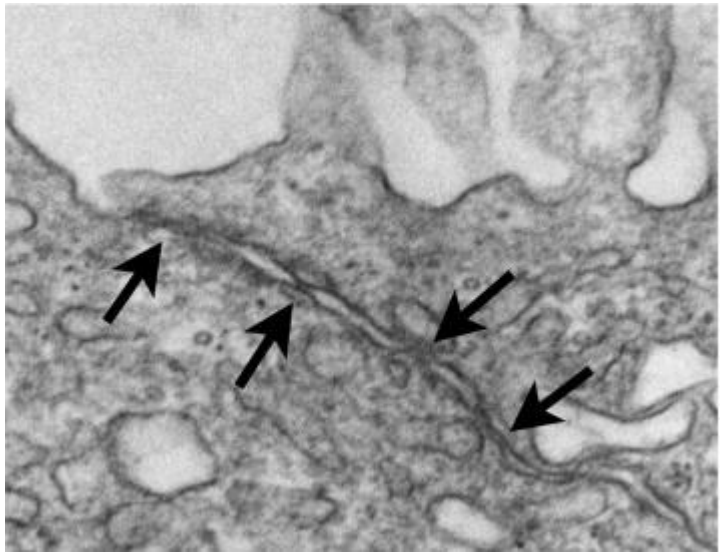


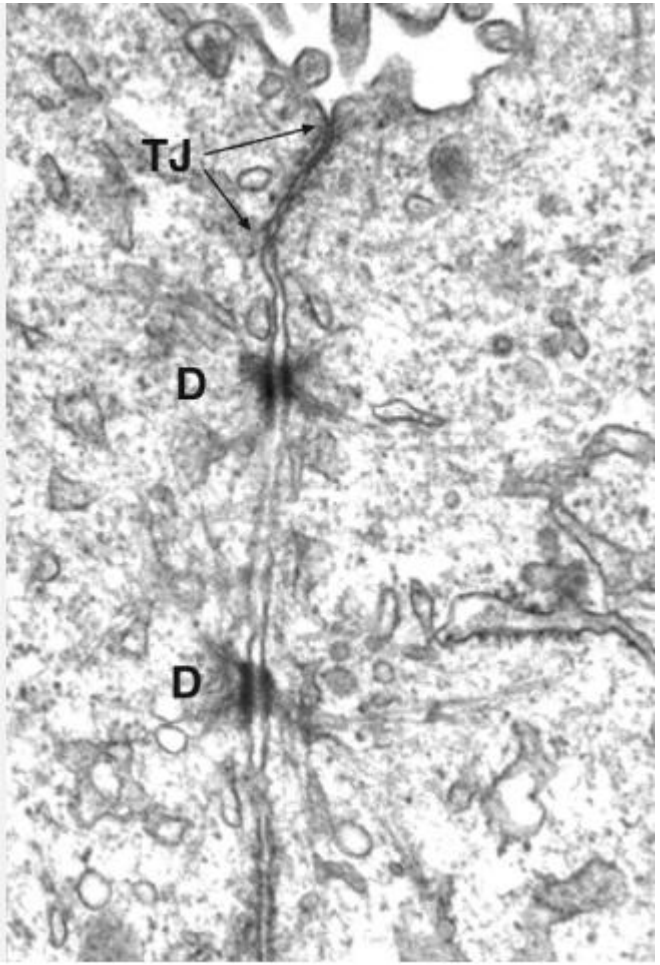
Figure 22.22
Molecular Cell Biology, eighth edition
Don W. Fawcett/T. Reese/Science Source.



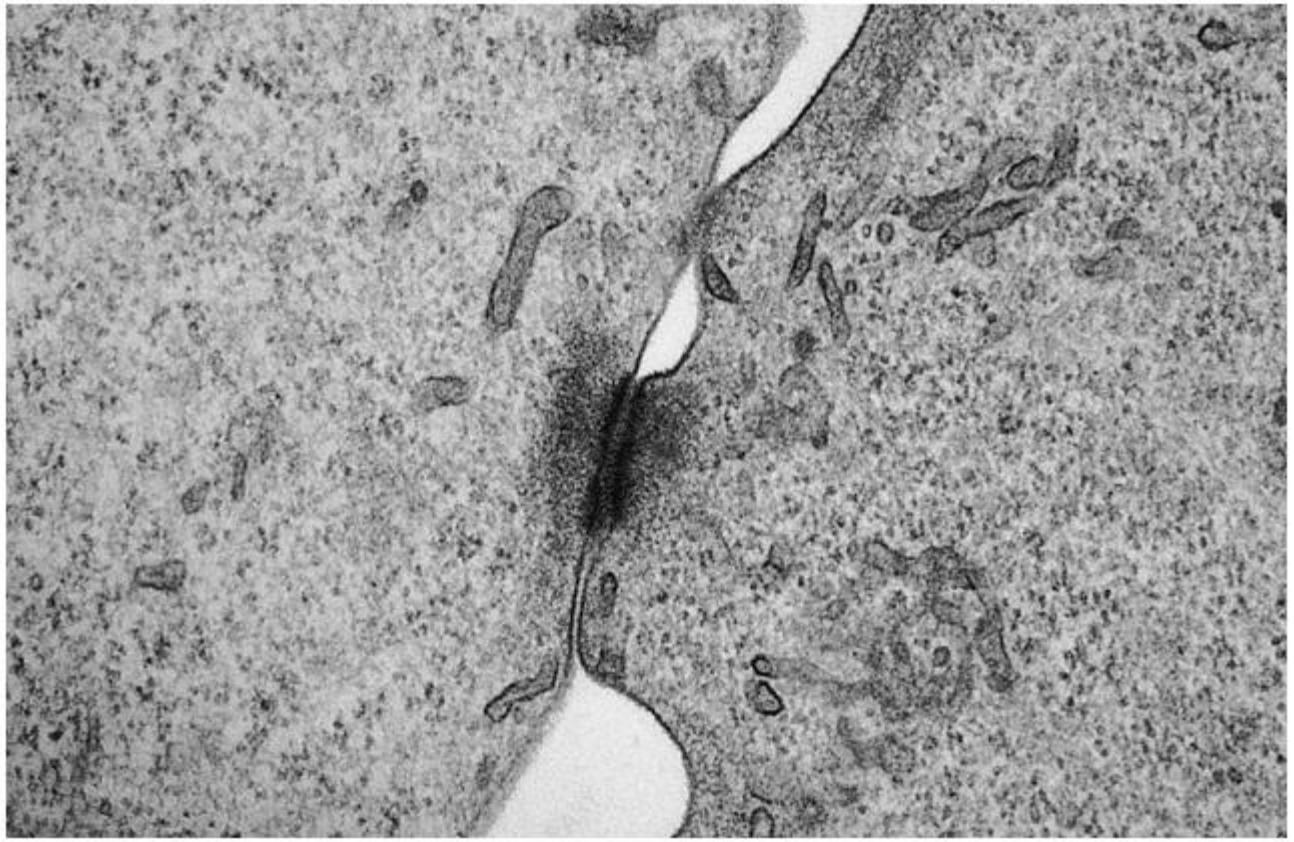
Complejos de unión







Desmosome

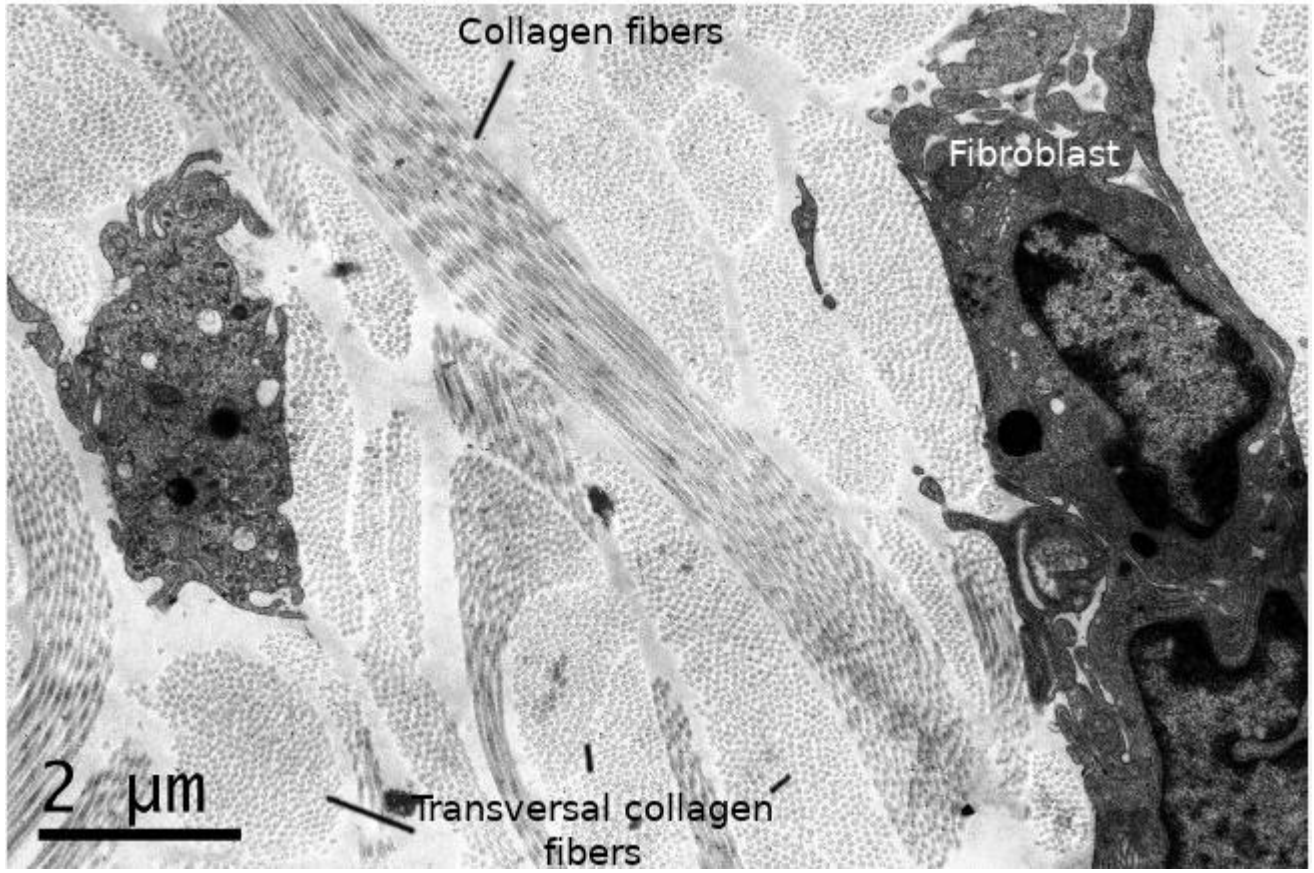


Credit: ANNE WESTON, EM STP, THE FRANCIS CRICK INSTITUTE / SCIENCE PHOTO LIBRARY

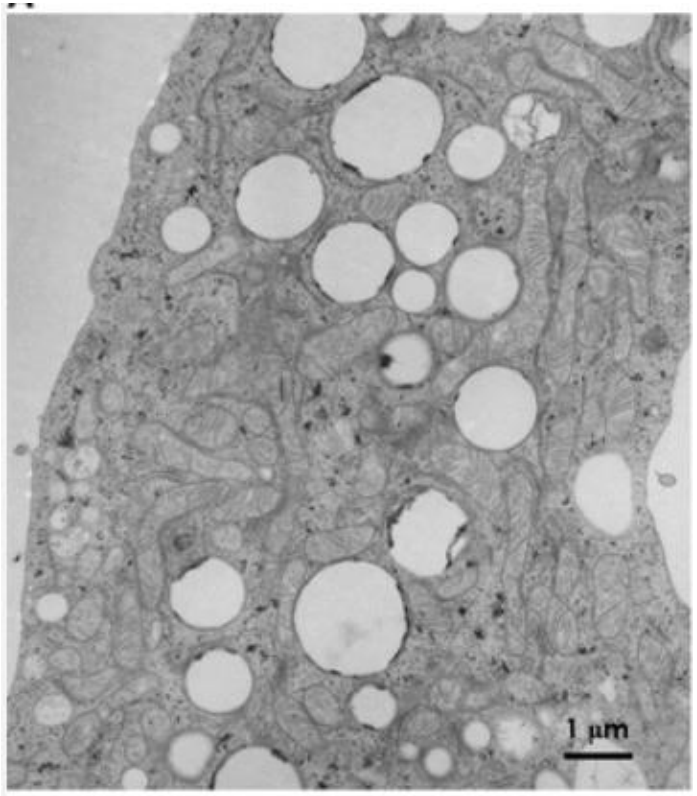
Restrictions: Images not to be used by the tobacco industry.

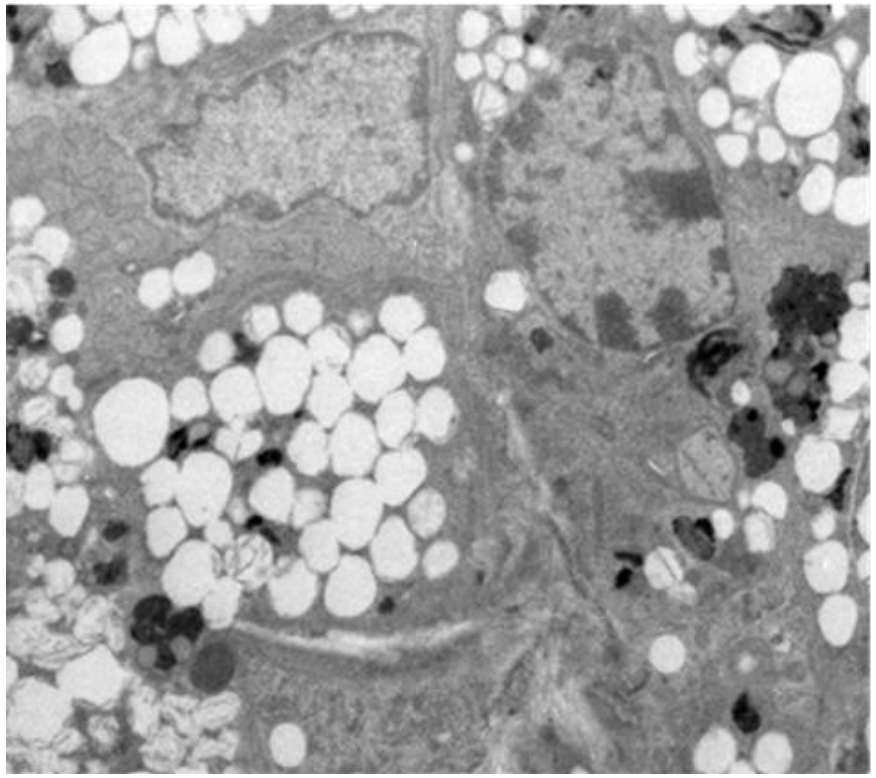
Caption: Desmosome (also known as macula adherens) between two cells, transmission electron micrograph (TEM). Desmosomes are intercellular junctions that provide strong adhesion between cells and their role is to bind adjacent cells together. Desmosomes have intermediate filaments that help anchor the junction. Faulty desmosome function can result in a number of conditions and diseases, particularly skin diseases and disorders

Espacio extracelular (colágeno etc)

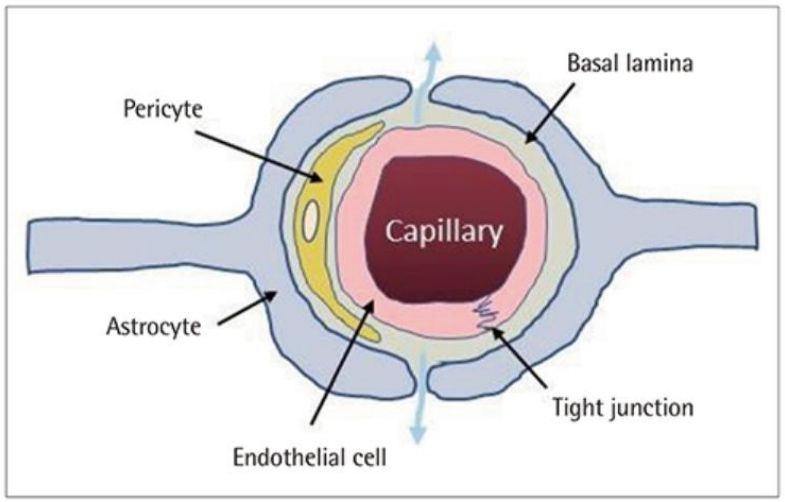


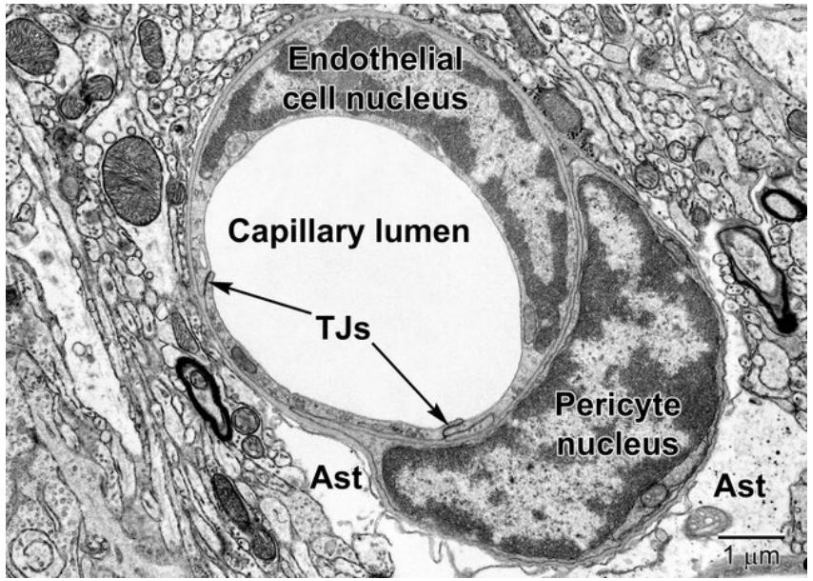
Gotas lipidicas





Capilares
(ej.cerebrales)





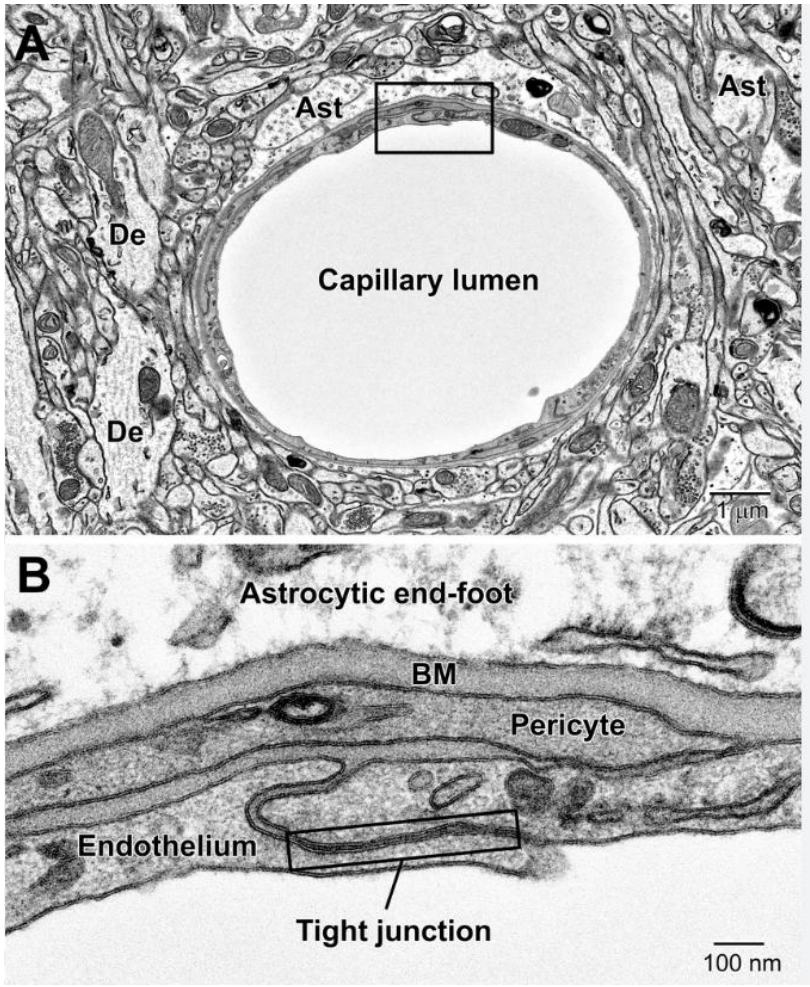


Fig. 1. The blood-brain barrier (BBB) is an important component of the neurovascular unit. BBB is the structural and functional unit formed by the endothelial cells, pericytes, basal lamina (or called basement membrane sometimes though the two terms differ to some extent), and the end feet of astrocytes.

Investig Magn Reson Imaging. 2022 Dec;26(4):229-236.
https://doi.org/10.13104/imri.2022.26.4.229

Brain Ultrastructure: Putting the Pieces Together

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DOI: 10.3389/fcell.2021.629503



Globulo rojo en un capilar

https://commons.wikimedia.org/wiki/File:A_red_blood_cell_in_a_capillary,_pancreatic_tissue_-_TEM.jpg

Para el cálculo del tamaño de las estructuras

- Referencia: microtúbulos tienen un diámetro (d)=25nm
- Distancia real: distancia que hemos medido en la foto/aumento
- Barra de calibración: usar la "regla de tres" usando la distancia conocida en la foto (ej. Obtenida basándose en el "d" de los microtubulos)